

Ruche training course

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école —
normale —
supérieure —
paris—saclay —



université
PARIS-SACLAY



- 1 Mésocentre Paris Saclay
- 2 Before computing
- 3 SLURM Workload manager
- 4 Examples
- 5 Visualization interface

Objectives of the Mesocenter

- Providing a set of IT resources and software dedicated to scientific computing
- Helping with their use
- Contributing to the pooling of expertise
- Conducting experiments with new computer technologies related to science

Platforms

- **Ruche**
- Lab-IA : GPUs only platform designed for AI
- Cloud@VirtualData : cloud computing + bare metal systems

The objectives of Ruche

Providing researchers with an efficient computational environment

- computational resources well-suited to the various needs (combustion, energy, biomathematics, mechanical, signal processing, electronic CAD, optimization, etc.)
- a software environment designed for code development (compiling, debugging and profiling tools)
- proximity support

Creating a research-oriented tool

- promoting access to HPC
- stimulating the development of new numerical methods and algorithms
- offering a place for scalability tests and improvement (needed for GENCI and PRACE access)

Personnel organization of **Ruche**

Support team

Human resources:

- 6 support engineers and 3 system administrators (3.2 FTE)

Tasks:

- machine administration
- management of the software environment
- user support
- project backing

Technical committee

Composition:

- scientific directors
- support team
- laboratories referents

Tasks:

- supervising the activity of **Ruche**
- defining the technical guidelines of **Ruche**
- dialogue with the users of the involved laboratories

Recent resources

- 2 connexion nodes
- GPFS file system (480 Tio, 8.6 Gio/s)
- Infiniband network 100 Gb/s
- Slurm scheduler
- Remote visualisation system
- 216 computation nodes with 2-CPU Intel Xeon "Cascade Lake" 6230 20c 2.1GHz 192GB
- 14 large memory nodes with 4-CPU Intel Xeon "Cascade Lake" 6230 20c 2.1GHz 1.5 TB
- 10 GPGPU nodes with 4 Nvidia Tesla V100 32GB
- 9 GPGPU nodes with 4 Nvidia Tesla A100 40GB

Resources from the Fusion supercomputer

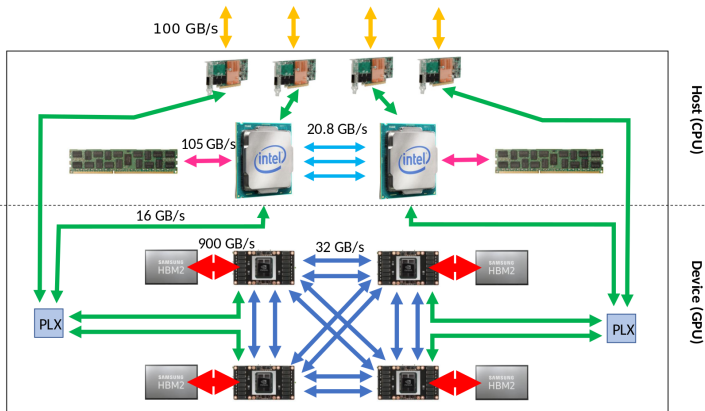
- 1 nodes with 4-CPU Intel Xeon "Skylake" 6148 20c 2.4GHz 1.5 TB
- 3 nodes with 4-CPU Intel Xeon "Skylake" 6148 20c 2.4GHz 768 GB
- 16 nodes with 2-CPU Intel Xeon "Skylake" 6148 20c 2.4GHz 192 GB
- 4 GPGPU nodes with 2 Nvidia Tesla P100 12GB

Ruche supercomputer, outline



Ruche supercomputer, node example

GPU node: 2 Intel Xeon Gold 6230 20C + 4 Nvidia Tesla V100



- General information on the mesocenter:

`https://mesocentre.universite-paris-saclay.fr/`

- Documentation:

`https://mesocentre.pages.centralesupelec.fr/user\_doc/`

- Contact the support team:

`ruche.support@universite-paris-saclay.fr`

Specify every information you have:

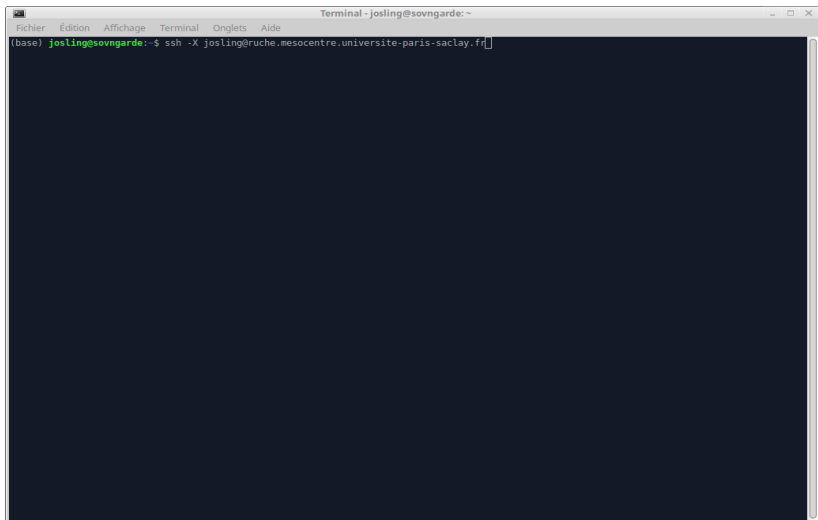
- * Login
- * Lab
- * Nature of the problem
- * Job's ID
- * Outputs
- * Permission to access your data

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 - File transfer
 - The module command
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 - Python environments
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On Linux, Mac and recent Windows

Ruche is accessible from everywhere using SSH.
X11 forwarding can be enabled.

A screenshot of a terminal window titled "Terminal - josling@sovngarde: ~". The window has a menu bar with "Fichier", "Édition", "Affichage", "Terminal", "Onglets", and "Aide". The terminal content shows a prompt "(base) josling@sovngarde:~\$" followed by the command "ssh -X josling@ruche.mesocentre.universite-paris-saclay.fr". The command is partially executed, with a cursor at the end. The terminal background is dark blue, and the text is light blue/green. A vertical scrollbar is visible on the right side of the terminal window.

```
Terminal - josling@sovngarde: ~  
Fichier  Édition  Affichage  Terminal  Onglets  Aide  
(base) josling@sovngarde:~$ ssh -X josling@ruche.mesocentre.universite-paris-saclay.fr
```


On Linux, Mac and recent Windows

```
Terminal - josling@ruche02:~
Fichier  Édition  Affichage  Terminal  Onglets  Aide
(base) josling@sovgarde:~$ ssh -X josling@ruche.mesocentre.universite-paris-saclay.fr
Warning: Permanently added the ECDSA host key for IP address '129.175.109.7' to the list of known hosts.
Last login: Thu Oct  7 16:22:18 2021

      Bonjour !

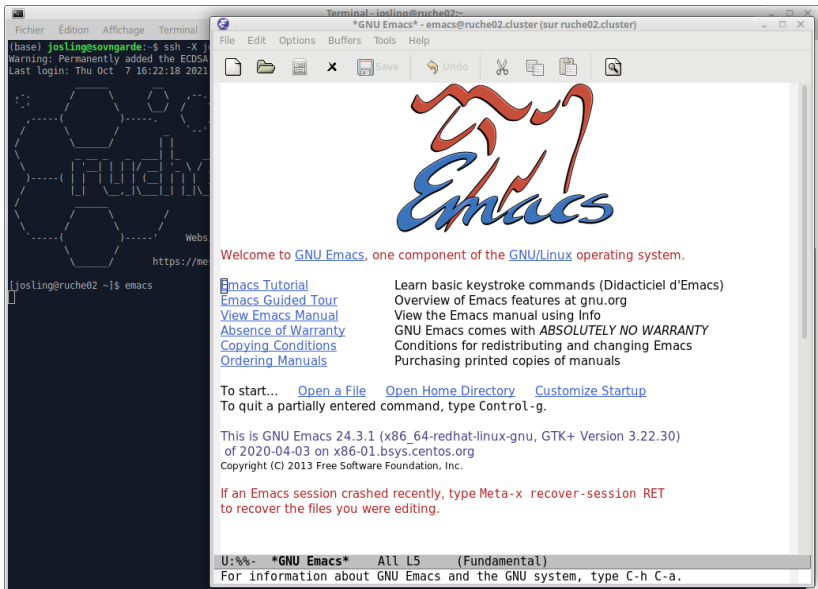
      the Mésocenter team wishes you
      a great time computing on ruché.

      We appreciate your feedback.
      -----

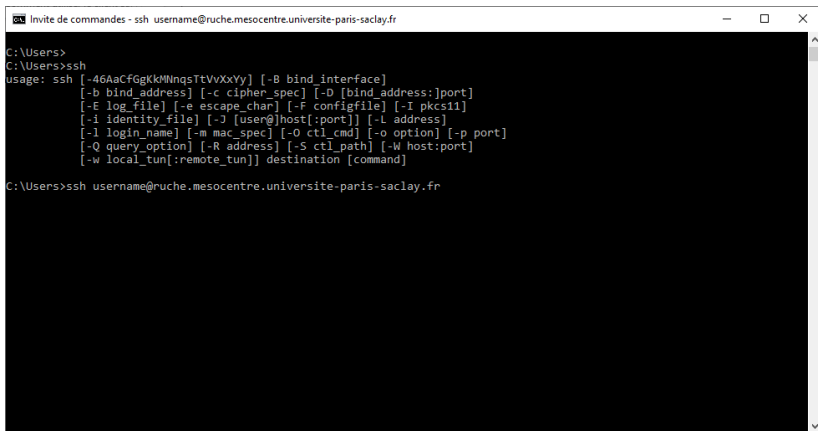
      Support : ruché_support@groupe.renater.fr
      Website : http://mesocentre.centralesupelec.fr/
               https://mesocentre.pages.centralesupelec.fr/user_doc/

[josling@ruche02 ~]$
```

On Linux, Mac and recent Windows



On Linux, Mac and recent Windows



```
Invite de commandes - ssh username@ruche.mesocentre.universite-paris-saclay.fr
C:\Users>
C:\Users>ssh
usage: ssh [-46AaCfGgKkMnQsTtVvXxYy] [-B bind_interface]
          [-b bind_address] [-c cipher_spec] [-D [bind_address:]port]
          [-E log_file] [-e escape_char] [-F configfile] [-I pkcs11]
          [-i identity_file] [-J [user@]host[:port]] [-L address]
          [-l login_name] [-m mac_spec] [-O ctl_cmd] [-o option] [-p port]
          [-Q query_option] [-R address] [-S ctl_path] [-W host:port]
          [-w local_tun[:remote_tun]] destination [command]

C:\Users>ssh username@ruche.mesocentre.universite-paris-saclay.fr
```

Using XQuartz on Mac

XQuartz <https://www.xquartz.org/> provides a version of the X.Org X Window System that runs on OS X.

The XQuartz project is an open-source effort to develop a version of the [X.Org X Window System](#) that runs on OS X. Together with supporting libraries and applications, it forms the X11.app that Apple shipped with OS X versions 10.5 through 10.7.

Quick Download

Download	Version	Released	Info
XQuartz-2.7.11.dmg	2.7.11	2016-10-29	For OS X 10.6.3 or later

License Info

An XQuartz installation consists of many individual pieces of software which have various licenses. The X.Org software components' licenses are discussed on the [X.Org Foundation Licenses](#) page. The [quartz-rem](#) window manager included with the XQuartz distribution uses the [Apple Public Source License Version 2.0](#).

Web site based on a design by John D. Minkley for the XQuartz project.
With site content distribution services provided by CloudFlare.

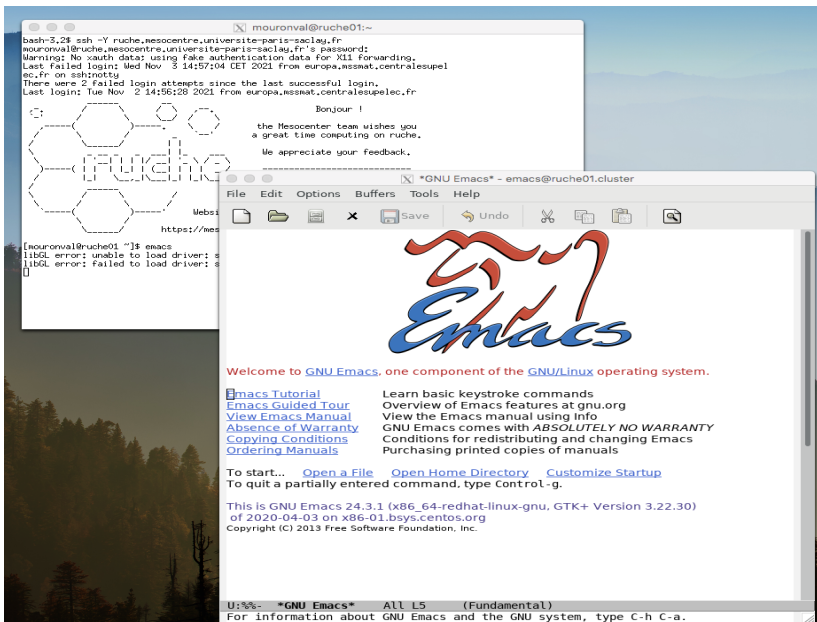
Distributed by
 Jfrog Bintray

Using XQuartz on Mac



```
bash-3.2$ ssh -Y ruche.mesocentre.universite-paris-saclay.fr
```

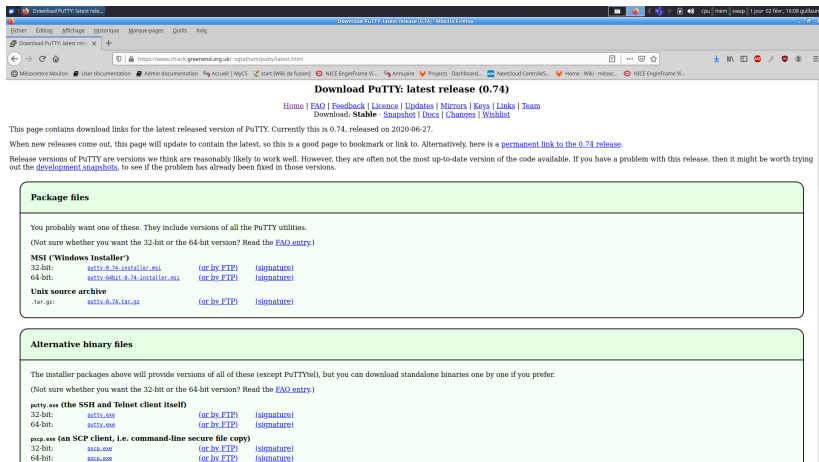
Using XQuartz on Mac



Using PuTTY on windows

To access Ruche from a Windows machine, a SSH client is mandatory.

PuTTY:: <https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html>



The screenshot shows a web browser displaying the PuTTY website. The page title is "Download PuTTY: latest release (0.74)". The browser's address bar shows the URL "https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html". The page content includes a navigation bar with links like Home, FAQ, Feedback, Licence, Updates, Mirrors, Keys, Links, and Team. Below the navigation bar, there is a section titled "Download PuTTY: latest release (0.74)" with a sub-header "Download: Stable - Snapshots | Docs | Changes | Wishlist". The main content area is divided into two sections: "Package files" and "Alternative binary files".

Download PuTTY: latest release (0.74)

Home | [FAQ](#) | [Feedback](#) | [Licence](#) | [Updates](#) | [Mirrors](#) | [Keys](#) | [Links](#) | [Team](#)
Download: [Stable](#) - [Snapshots](#) | [Docs](#) | [Changes](#) | [Wishlist](#)

This page contains download links for the latest released version of PuTTY. Currently this is 0.74, released on 2020-06-27.

When new releases come out, this page will update to contain the latest, so this is a good page to bookmark or link to. Alternatively, here is a [permanent link to the 0.74 release](#).

Release versions of PuTTY are versions we think are reasonably likely to work well. However, they are often not the most up-to-date version of the code available. If you have a problem with this release, then it might be worth trying out the [development snapshots](#), to see if the problem has already been fixed in those versions.

Package files

You probably want one of these. They include versions of all the PuTTY utilities.
(Not sure whether you want the 32-bit or the 64-bit version? Read the [FAQ entry](#).)

MSI ("Windows Installer")

32-bit:	putty-0.74-installer.msi	(or by FTP)	(signature)
64-bit:	putty-64bit-0.74-installer.msi	(or by FTP)	(signature)

Unix source archive

.tar.gz:	putty-0.74.tar.gz	(or by FTP)	(signature)
----------	-----------------------------------	-------------------------------	-------------------------------

Alternative binary files

The installer packages above will provide versions of all of these (except PuTTYtel), but you can download standalone binaries one by one if you prefer.
(Not sure whether you want the 32-bit or the 64-bit version? Read the [FAQ entry](#).)

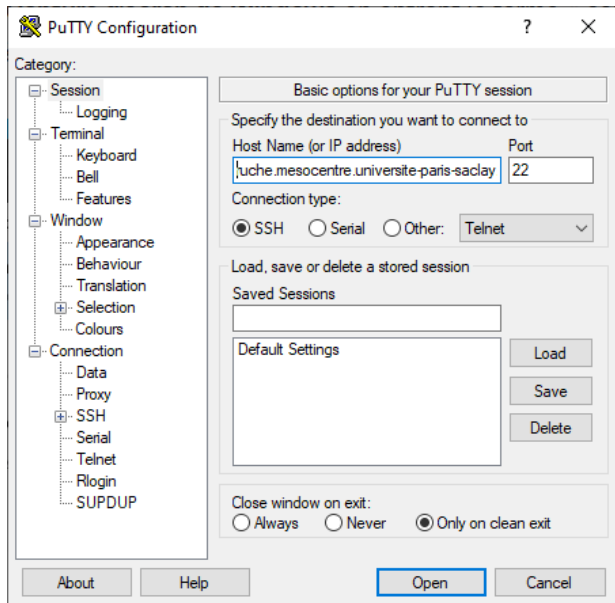
putty.exe (the SSH and Telnet client itself)

32-bit:	putty.exe	(or by FTP)	(signature)
64-bit:	putty.exe	(or by FTP)	(signature)

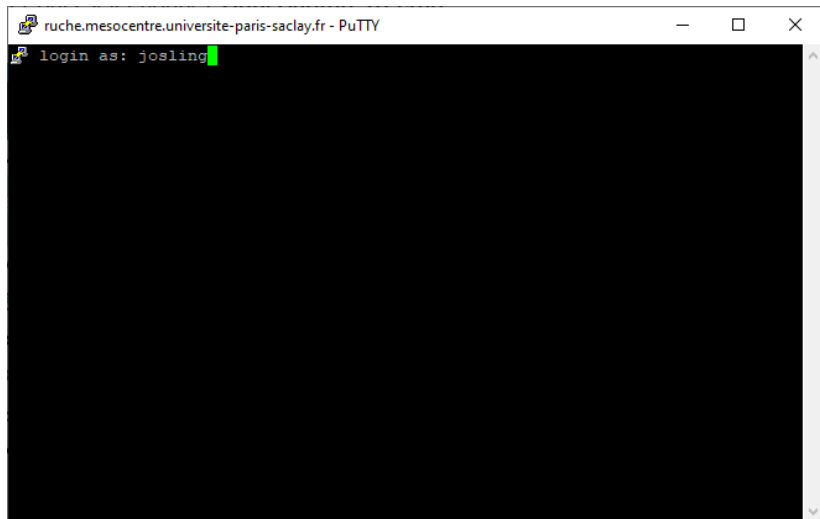
pscp.exe (an SCP client, i.e. command-line secure file copy)

32-bit:	pscp.exe	(or by FTP)	(signature)
64-bit:	pscp.exe	(or by FTP)	(signature)

Using PuTTY on windows



Using Putty on windows



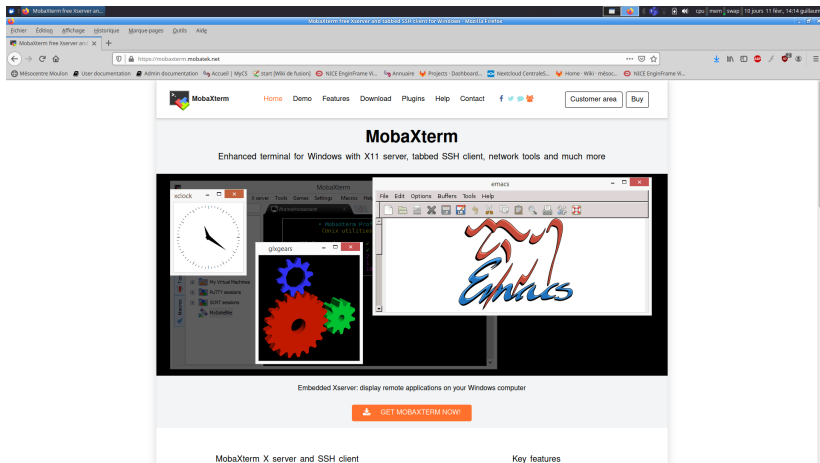
Using Putty on windows

```
josling@ruche02:~  
login as: josling  
josling@ruche.mesocentre.universite-paris-saclay.fr's password:  
Last failed login: Wed Nov  3 16:06:11 CET 2021 from 138.195.215.194 on ssh:notty  
There was 1 failed login attempt since the last successful login.  
Last login: Wed Nov  3 15:51:02 2021 from 138.195.215.188  
  
      Bonjour !  
  
the Mésocenter team wishes you  
a great time computing on ruche.  
  
We appreciate your feedback.  
-----  
  
Support : ruche\_support@groupes.renater.fr  
Website : http://mesocentre.centralesupelec.fr/  
https://mesocentre.pages.centralesupelec.fr/user\_doc/  
[josling@ruche02 ~]$
```

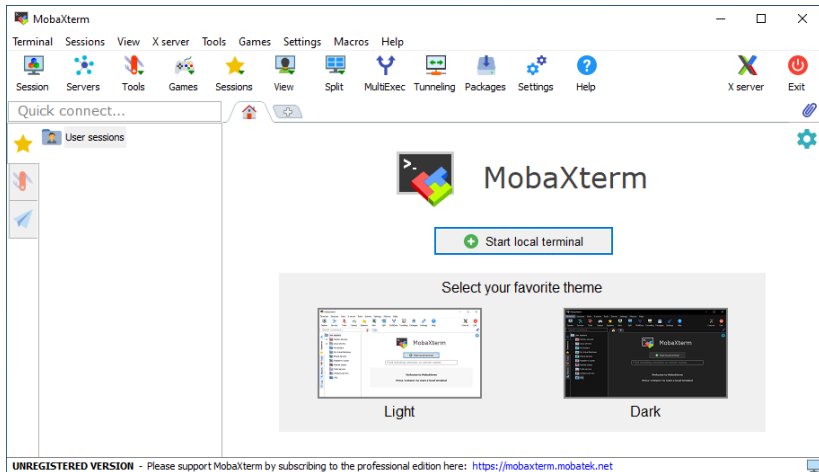
Using MobaXterm on windows

To access **Ruche** from a Windows machine, a SSH client is mandatory.

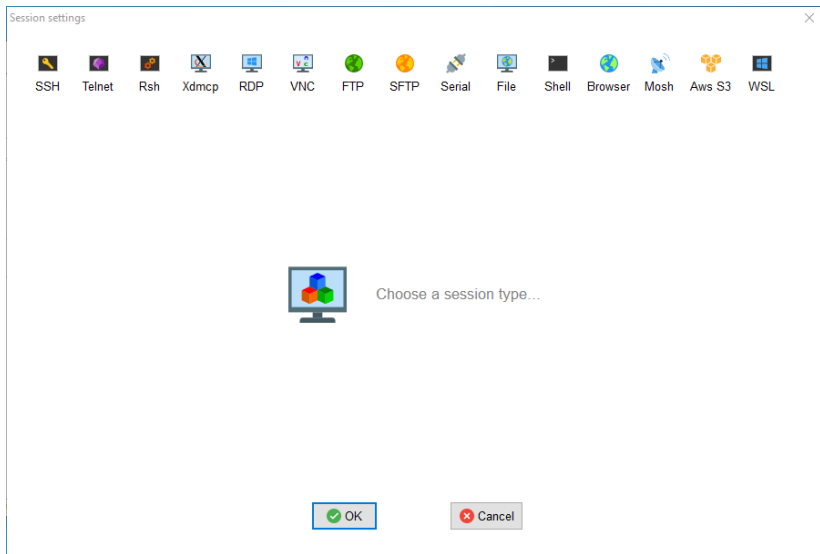
MobaXterm:: <https://mobaxterm.mobatek.net/>



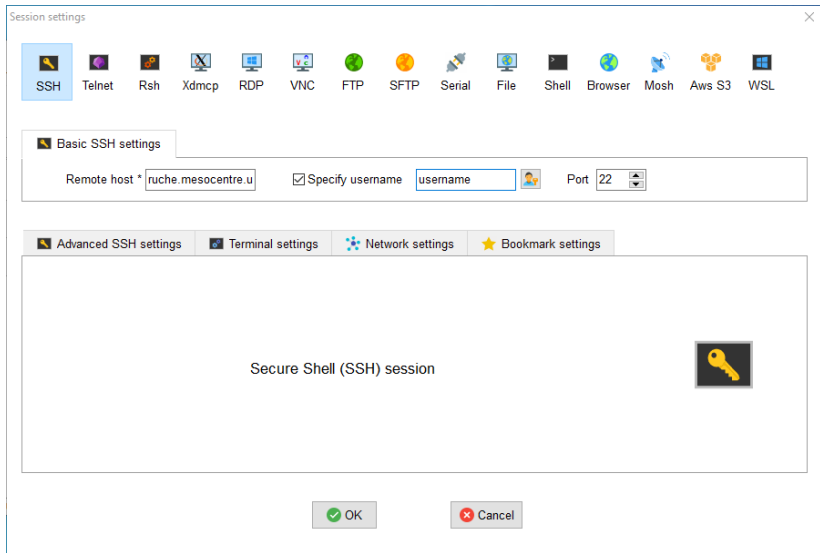
Using MobaXterm on windows



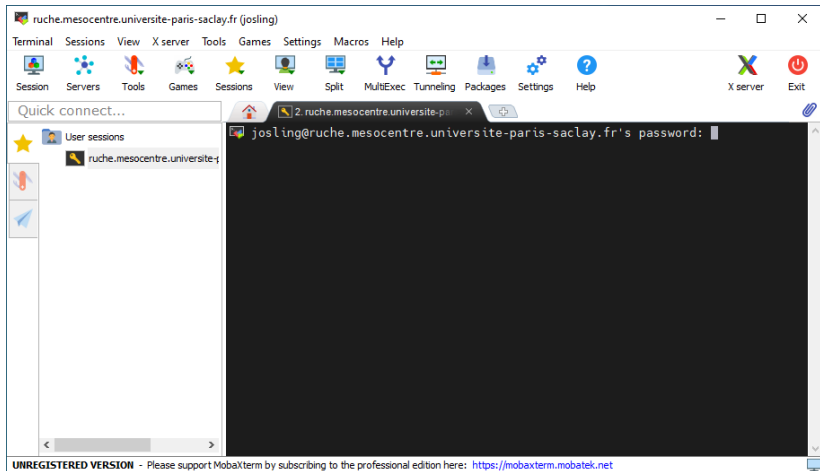
Using MobaXterm on windows



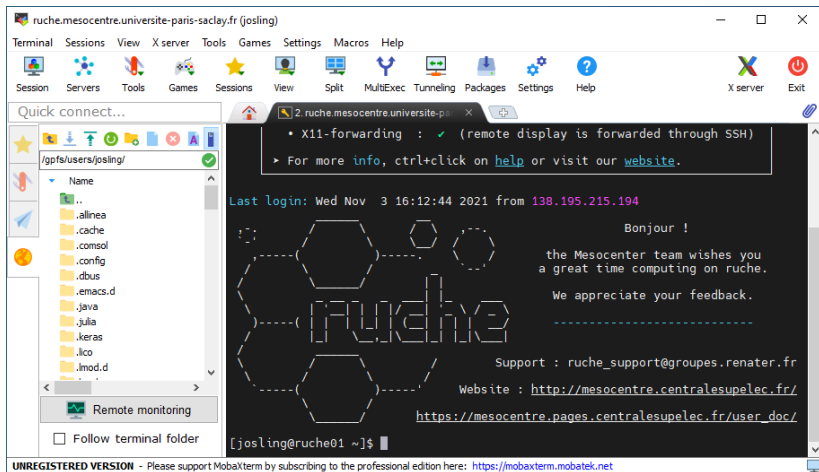
Using MobaXterm on windows



Using MobaXterm on windows



Using MobaXterm on windows



- Use strong passwords (long, outside of a dictionary, with complex signs)
- Use a different password for each service (do not use your institutional password)
- Protect your passwords (no sharing, no file on desktop, no post-it note)
- Do not use personal information in your passwords
- Change the default password

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Ruche support team does not provide support in using Linux.

English

- <https://cheatography.com/davechild/cheat-sheets/linux-command-line/>

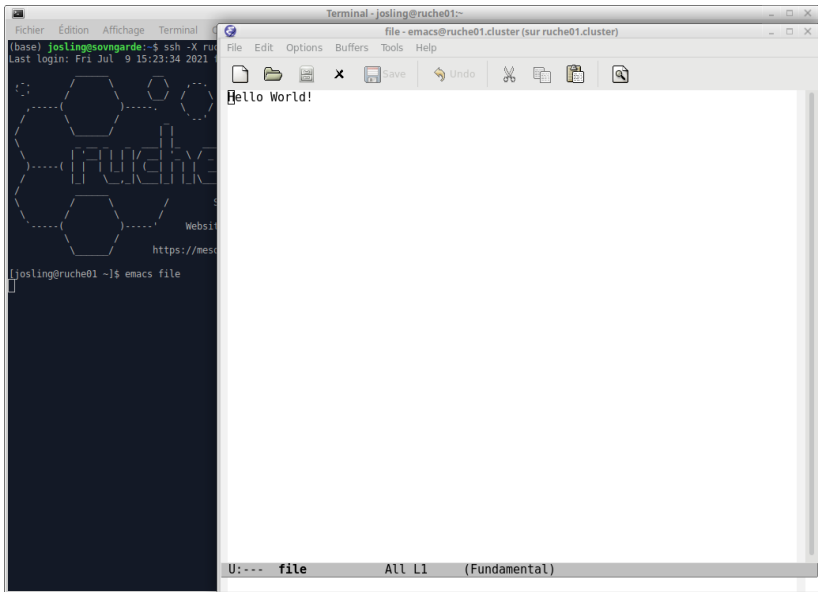
French

- https://doc.ubuntu-fr.org/tutoriel/console_ligne_de_commande
- <http://juliend.github.io/linux-cheatsheet/>

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```
Terminal - josling@ruche01:~  
Fichier  Édition  Affichage  Terminal  Onglets  Aide  
(base) josling@sovgarde:~$ ssh -X ruche  
Last login: Fri Jul  9 15:23:34 2021 from lfbn-idf2-1-419-217.w86-246.abo.wanadoo.fr  
  
      Bonjour !  
  
      the Mesocenter team wishes you  
      a great time computing on ruche.  
  
      We appreciate your feedback.  
      -----  
  
      Support : ruche_support@groupe.renater.fr  
      Website : http://mesocentre.centralesupelec.fr/  
               https://mesocentre.pages.centralesupelec.fr/user\_doc/  
  
[josling@ruche01 ~]$ emacs file  
[
```

Emacs



```
Terminal - josling@ruche01:~
Fichier  Édition  Affichage  Terminal  Onglets  Aide
(base) josling@ovngarde:~$ ssh -X ruche
Last login: Fri Jul 9 15:26:02 2021 from lfbn-idf2-1-419-217.w86-246.abo.wanadoo.fr

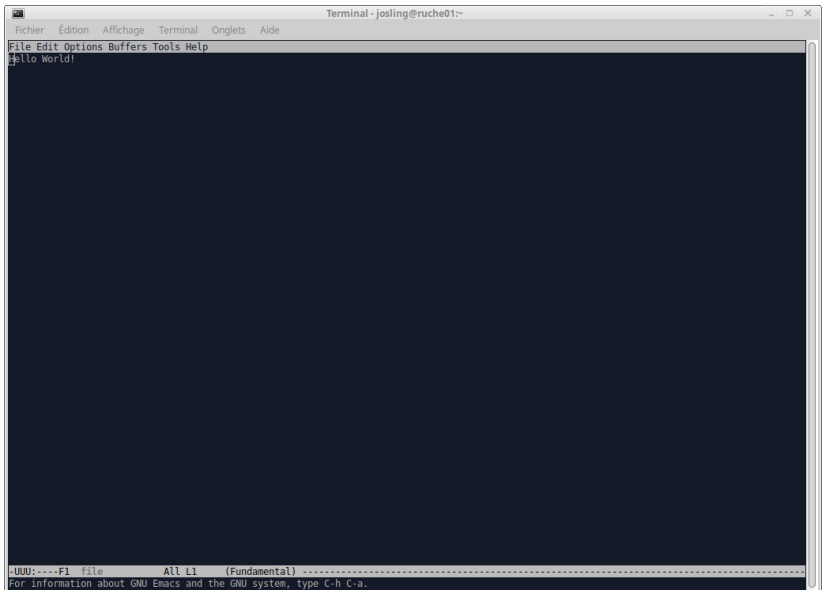
      Bonjour !

      the Mesocenter team wishes you
      a great time computing on ruche.

      We appreciate your feedback.
      -----

      Support : ruche_support@groupes.renater.fr
      Website : http://mesocentre.centralesupelec.fr/
               https://mesocentre.pages.centralesupelec.fr/user\_doc/

[josling@ruche01 ~]$ emacs -nw file
```




```

Terminal - josling@ruche01:~
Fichier  Édition  Affichage  Terminal  Onglets  Aide
(base) josling@sovgarde:~$ ssh -X ruché
Last login: Fri Jul  9 15:19:40 2021 from lfbn-idf2-1-419-217.w86-246.abo.wanadoo.fr

  Bonjour !

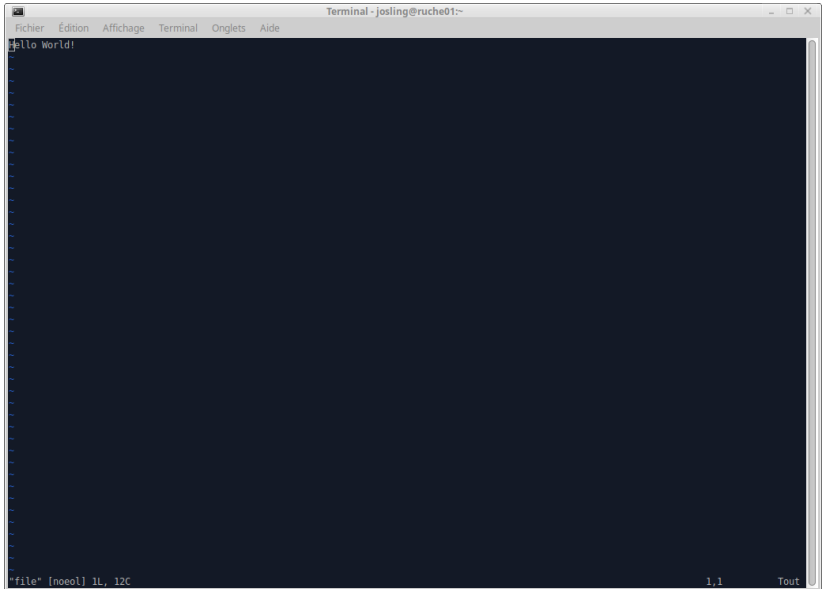
the Mesocenter team wishes you
a great time computing on ruché.

We appreciate your feedback.
-----

Support : ruché_support@groupe.s.renater.fr
Website : http://mesocentre.centralesupelec.fr/
https://mesocentre.pages.centralesupelec.fr/user_doc/

[josling@ruche01 ~]$ vim file

```



```

Terminal - josling@ruche01:~
Fichier  Édition  Affichage  Terminal  Onglets  Aide
(base) josling@sovgarde:~$ ssh -X ruche
Last login: Fri Jul  9 15:22:22 2021 from lfbn-idf2-1-419-217.w86-246.abo.wanadoo.fr

  Bonjour !

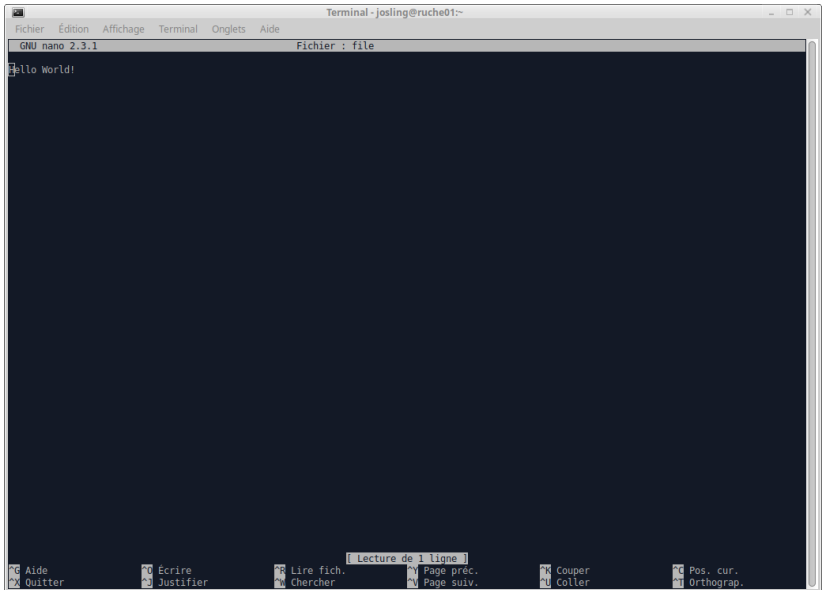
the Mesocenter team wishes you
a great time computing on ruche.

We appreciate your feedback.
-----

Support : ruche_support@groupes.renater.fr
Website : http://mesocentre.centralesupelec.fr/
https://mesocentre.pages.centralesupelec.fr/user_doc/

[josling@ruche01 ~]$ nano file

```



The image shows a terminal window titled "Terminal - josling@ruche01:~". Inside the terminal, the GNU nano 2.3.1 text editor is open, editing a file named "file". The editor's interface includes a menu bar at the top with options: Fichier, Édition, Affichage, Terminal, Onglets, and Aide. The main editing area is dark blue with white text, showing the line "Hello World!". At the bottom, a status bar displays various keyboard shortcuts for navigation and editing, such as ^G Aide, ^O Écrire, ^R Lire fich., ^Y Page préc., ^X Couper, ^C Pos. cur., ^Q Quitter, ^J Justifier, ^W Chercher, ^V Page suiv., and ^U Coller. A small status indicator in the center of the bottom bar reads "[Lecture de 1 ligne]".

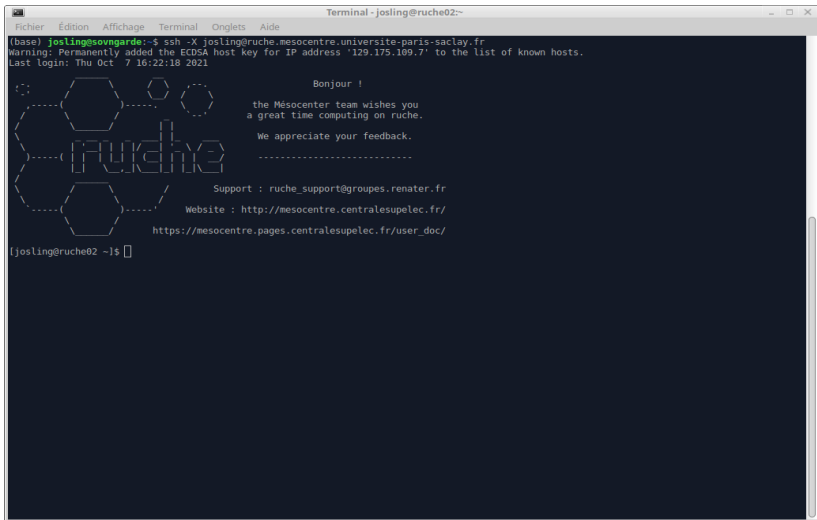
```
Terminal - josling@ruche01:~
Fichier  Édition  Affichage  Terminal  Onglets  Aide
GNU nano 2.3.1                                Fichier : file
Hello World!

^G Aide    ^O Écrire  ^R Lire fich.  ^Y Page préc.  ^X Couper  ^C Pos. cur.
^Q Quitter ^J Justifier ^W Chercher    ^V Page suiv.  ^U Coller  ^T Orthograp.
[ Lecture de 1 ligne ]
```

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Front-end nodes

When the SSH tunnel is established, you are connected to one of **Ruche** 's front-end nodes, where you can edit your source files and compile your code.
You must not run codes on these nodes.



```
Terminal - josling@ruche02:~  
Fichier  Édition  Affichage  Terminal  Onglets  Aide  
(base) josling@sovgarde:~$ ssh -X josling@ruche.mesocentre.universite-paris-saclay.fr  
Warning: Permanently added the ECDSA host key for IP address '129.175.109.7' to the list of known hosts.  
Last login: Thu Oct  7 16:22:18 2021  
  
      Bonjour !  
  
      the Mésocenter team wishes you  
      a great time computing on ruche.  
  
      We appreciate your feedback.  
      -----  
  
      Support : ruche_support@groupes.renater.fr  
      Website : http://mesocentre.centralesupelec.fr/  
               https://mesocentre.pages.centralesupelec.fr/user\_doc/  
  
[josling@ruche02 ~]$
```

Warning message

Dear username,

This is an automatic warning email from ruche cluster.

A process associated with your account on ruche login node ruche02 has been detected and killed:
[username@ruche01]\$./prog_seq.exe

This process has been stopped for being too long and CPU demanding. Such processes prevent all other users.

If this process is a resource demanding code, please use the batch scheduler to run it on a proper computer.

```
#!/bin/bash
```

```
# First part : Slurm directives to reserve nodes, memory and time
```

```
#SBATCH --job-name=hello_seq
```

```
#SBATCH --output=%x.o%j
```

```
#SBATCH --time=00:20:00
```

```
#SBATCH --ntasks=1
```

```
#SBATCH --partition=cpu_short
```

```
# Second part : Unix script to execute the code
```

```
# Load necessary modules
```

```
# Run executable
```

```
./prog_seq.exe
```

Then you can submit the job with the sbatch command in order to execute the code on computing nodes:
[username@ruche01]\$ sbatch slurm_script.sh

The online documentation is available on our web site :

https://mesocentre.pages.centralesupelec.fr/user_doc/ruche/06_slurm_jobs_management/

Thanks in advance.

Sincerely yours.

L'équipe support de Ruche

Personal directories

As a user, you have access to two separate directories for your own usage :

The users directory (or home directory or homedir)

Path: `/home/username`

also accessible via `/gpfs/users/username`

where you can store your source files (code, light data).

This volume is limited to 50 Go per user.

You can refer to this directory with the environment variable `$HOME`.

The working directory (or workdir)

Path: `/workdir/username`

also accessible via `/gpfs/workdir/username`

where you can store your binaries and data.

This volume is limited to 500 Go per user.

You can refer to this directory with the environment variable `$WORKDIR`.

**Neither your home directory nor your working directory are backed up.
Remember to recover all data before the closing of your account.**

You can quickly check your file quota with the command *ruche-quota*.

```
[username@ruche01 ~]$ ruche-quota
Disk usage on home :
  quota:  5,2G      / 50G          (10%)
  files:  66 267   / 200 000      (33%)
Disk usage on workdir :
  quota:  282G      / 1,1T         (26%)
  files:  1 910 782 / 8 000 000    (23%)
```

If your computations require more space on your *\$WORKDIR* directory, you can ask the support team for quota extension.

If your data is shared between members of your project and/or public (for example AI datasets), you can ask the support team for a shared directory.

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From Linux, Mac and recent Windows

scp: copying a file to Ruche

```
Terminal - josling@ruche01:~
Fichier  Édition  Affichage  Terminal  Onglets  Aide
(base) joslingsovgarde:~/my_directory$ more my_file
my_content
(base) joslingsovgarde:~/my_directory$ scp my_file josling@ruche.mesocentre.universite-paris-saclay.fr:/gpfs/users/josling/.
my_file                                     100% 10 1.0KB/s 00:00
(base) joslingsovgarde:~/my_directory$ ssh josling@ruche.mesocentre.universite-paris-saclay.fr
Last login: Wed Nov 3 17:00:36 2021

      Bonjour !

the Mesocenter team wishes you
a great time computing on ruche.

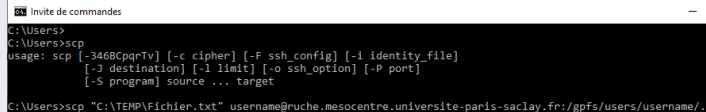
We appreciate your feedback.
-----

Support : ruche_support@groupes.renater.fr
Website : http://mesocentre.centralesupelec.fr/
https://mesocentre.pages.centralesupelec.fr/user_doc/

[josling@ruche01 ~]$ more my_file
my_content
[josling@ruche01 ~]$
```

From Linux, Mac and recent Windows

scp: copying a file to Ruche

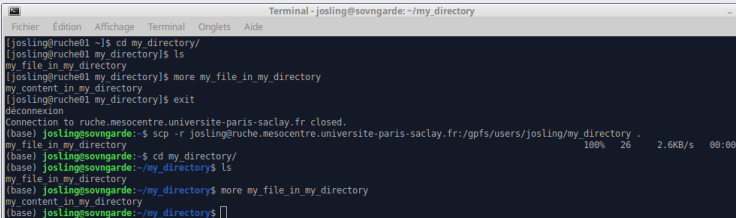


```
Invite de commandes
C:\Users>
C:\Users>scp
usage: scp [-3468CpqrTv] [-c cipher] [-F ssh_config] [-i identity_file]
          [-J destination] [-l limit] [-o ssh_option] [-P port]
          [-S program] source ... target

C:\Users>scp "C:\TEMP\Fichier.txt" username@ruche.mesocentre.universite-paris-saclay.fr:/gpfs/users/username/.
```

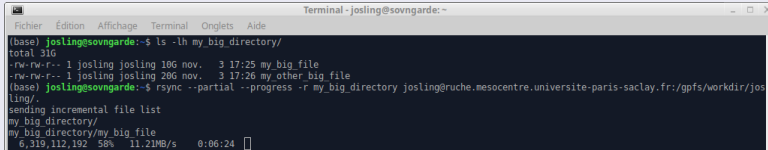
From Linux, Mac and recent Windows

scp: copying a directory from Ruche

A terminal window titled "Terminal - josling@sovngarde: ~/my_directory" with a menu bar (Fichier, Édition, Affichage, Terminal, Onglets, Aide). The terminal shows a sequence of commands and their outputs. The user 'josling' is at 'ruche01' and navigates to 'my_directory'. They use 'ls' to list files, 'more' to view 'my_file_in_my_directory', and 'exit' to leave. A message indicates the connection to 'ruche.mesocentre.universite-paris-saclay.fr' is closed. Then, the user 'joslingesovngarde' runs 'scp -r josling@ruche.mesocentre.universite-paris-saclay.fr:/gpfs/users/josling/my_directory .' from the 'base' prompt. The output shows '100% 26 2.6KB/s 00:00'. Finally, the user returns to 'my_directory' and repeats the 'ls', 'more', and 'exit' sequence.

```
Terminal - josling@sovngarde: ~/my_directory
Fichier  Édition  Affichage  Terminal  Onglets  Aide
[josling@ruche01 ~]$ cd my_directory/
[josling@ruche01 my_directory]$ ls
my_file_in_my_directory
[josling@ruche01 my_directory]$ more my_file_in_my_directory
my_content_in_my_directory
[josling@ruche01 my_directory]$ exit
déconnexion
Connection to ruche.mesocentre.universite-paris-saclay.fr closed.
(base) joslingesovngarde:~$ scp -r josling@ruche.mesocentre.universite-paris-saclay.fr:/gpfs/users/josling/my_directory .
my_file_in_my_directory                               100% 26    2.6KB/s   00:00
(base) joslingesovngarde:~$ cd my_directory/
(base) joslingesovngarde:~/my_directory$ ls
my_file_in_my_directory
(base) joslingesovngarde:~/my_directory$ more my_file_in_my_directory
my_content_in_my_directory
(base) joslingesovngarde:~/my_directory$
```

rsync: copying a large directory to Ruche (I)

A terminal window titled "Terminal - josling@sovngarde: ~" with a menu bar (Fichier, Édition, Affichage, Terminal, Onglets, Aide). The terminal shows the execution of an rsync command to copy a directory to a remote host. The output shows the directory listing and the progress of the rsync operation.

```
(base) josling@sovngarde:~$ ls -lh my_big_directory/
total 31G
-rw-rw-r-- 1 josling josling 10G nov.  3 17:25 my_big_file
-rw-rw-r-- 1 josling josling 20G nov.  3 17:26 my_other_big_file
(base) josling@sovngarde:~$ rsync --partial --progress -r my_big_directory josling@ruche.mesocentre.universite-paris-saclay.fr:/gpfs/workdir/josling/.
sending incremental file list
my_big_directory/
my_big_directory/my_big_file
 6,319,112,192 58% 11.21MB/s  0:06:24 [ ]
```

rsync: copying a large directory to Ruche (II)

```
Terminal - josling@sovngarde: ~
Fichier  Édition  Affichage  Terminal  Onglets  Aide
(base) josling@sovngarde:~$ ls -lh my_big_directory/
total 31G
-rw-rw-r-- 1 josling josling 10G nov.   3 17:25 my_big_file
-rw-rw-r-- 1 josling josling 20G nov.   3 17:26 my_other_big_file
(base) josling@sovngarde:~$ rsync --partial --progress -r my_big_directory josling@ruche.mesocentre.universite-paris-saclay.fr:/gpfs/workdir/josling/
sending incremental file list
my_big_directory/
my_big_directory/my_big_file
 6,613,073,920 61% 11.20MB/s   0:05:59 ^C
rsync error: received SIGINT, SIGTERM, or SIGHUP (code 20) at rsync.c(644) [sender=3.1.3]
rsync: [sender] write error: Broken pipe (32)
(base) josling@sovngarde:~$ rsync --partial --progress -r my_big_directory josling@ruche.mesocentre.universite-paris-saclay.fr:/gpfs/workdir/josling/
sending incremental file list
my_big_directory/my_big_file
10,737,418,240 100% 308.83MB/s   0:00:33 (xfr#1, to-chk=1/3)
my_big_directory/my_other_big_file
 7,824,277,504 36% 11.21MB/s   0:19:49 [
```

rsync: copying a large directory to Ruche (III)

```
Terminal - josling@ruche01/gpfs/workdir/josling
Fichier  Édition  Affichage  Terminal  Onglets  Aide
(base) josling@ruche01:~$ ls -lh my_big_directory/
total 31G
-rw-rw-r-- 1 josling josling 10G nov.  3 17:25 my_big_file
-rw-rw-r-- 1 josling josling 20G nov.  3 17:26 my_other_big_file
(base) josling@ruche01:~$ rsync --partial --progress -r my_big_directory josling@ruche.mesocentre.universite-paris-saclay.fr:/gpfs/workdir/josling/
sending incremental file list
my_big_directory/
my_big_directory/my_big_file
 6,613,073,920 61% 11.20MB/s  0:05:59 ^C
rsync error: received SIGINT, SIGTERM, or SIGHUP (code 20) at rsync.c(644) [sender=3.1.3]
rsync: [sender] write error: Broken pipe (32)
(base) josling@ruche01:~$ rsync --partial --progress -r my_big_directory josling@ruche.mesocentre.universite-paris-saclay.fr:/gpfs/workdir/josling/
sending incremental file list
my_big_directory/my_big_file
10,737,418,240 100% 308.83MB/s  0:00:33 (xfr#1, to-chk=1/3)
my_big_directory/my_other_big_file
21,474,836,480 100% 10.23MB/s  0:33:22 (xfr#2, to-chk=0/3)
(base) josling@ruche01:~$ ssh josling@ruche.mesocentre.universite-paris-saclay.fr
Last login: Wed Nov  3 17:31:25 2021

      Bonjour !

the Mesocenter team wishes you
a great time computing on ruche.

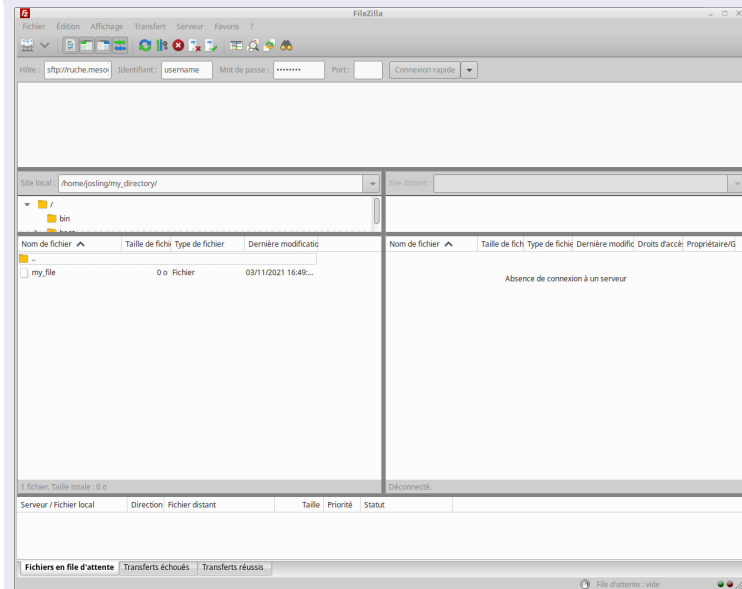
We appreciate your feedback.
-----

Support : ruche_support@groupe.renater.fr
Website : http://mesocentre.centralesupelec.fr/
https://mesocentre.pages.centralesupelec.fr/user_doc/

[josling@ruche01 ~]$ cd $WORKDIR/
[josling@ruche01 josling]$ ls -lh my_big_directory/
total 30G
-rw-r--r-- 1 josling mics 10G  3 nov.  17:41 my_big_file
-rw-r--r-- 1 josling mics 20G  3 nov.  18:14 my_other_big_file
[josling@ruche01 josling]$
```

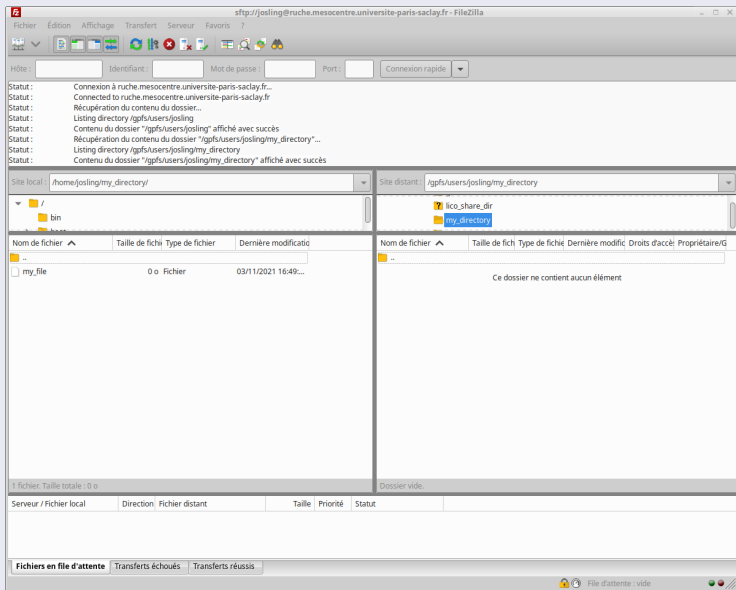

SFTP graphical clients: filezilla

Configuration



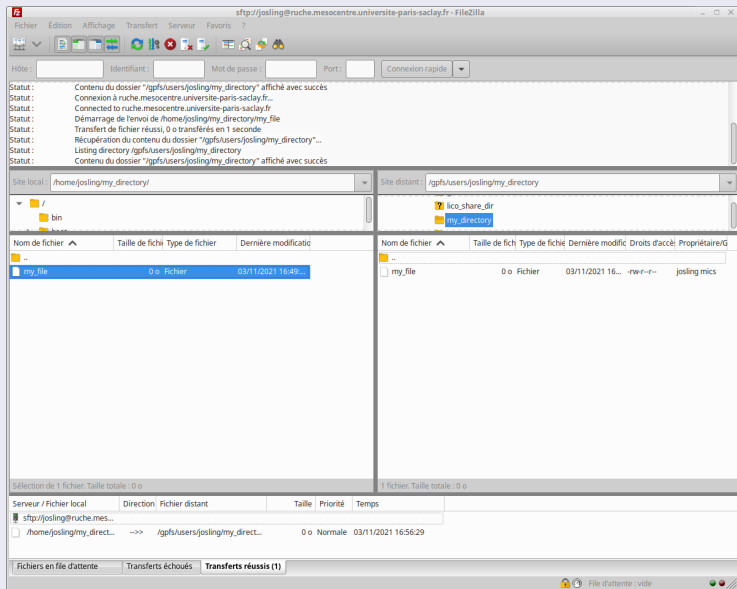
SFTP graphical clients: filezilla

Connection

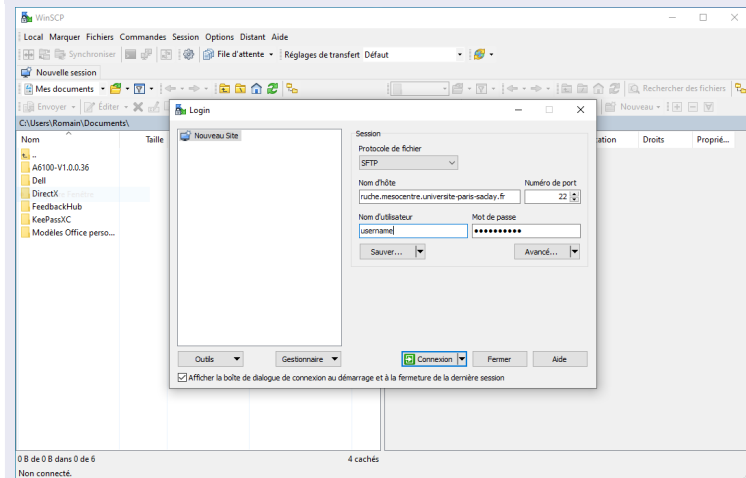


SFTP graphical clients: filezilla

Transfer

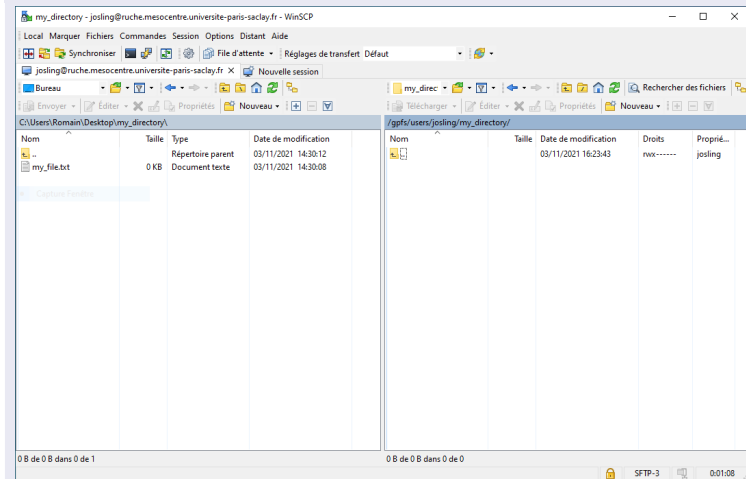


Configuration

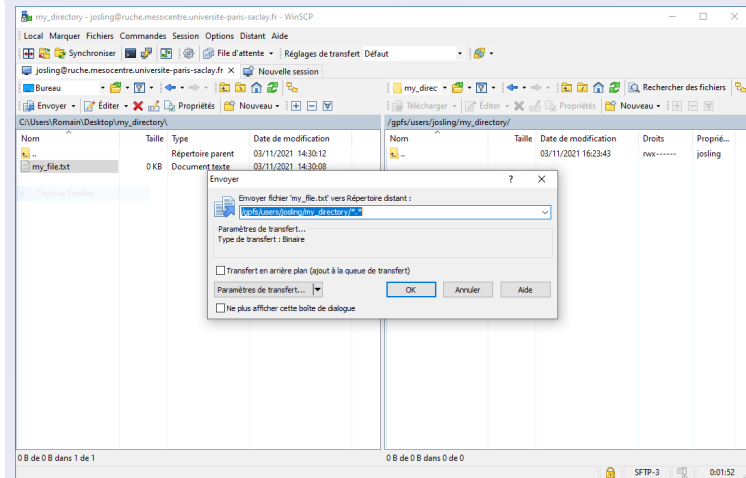


SFTP graphical clients: WinSCP

Connexion

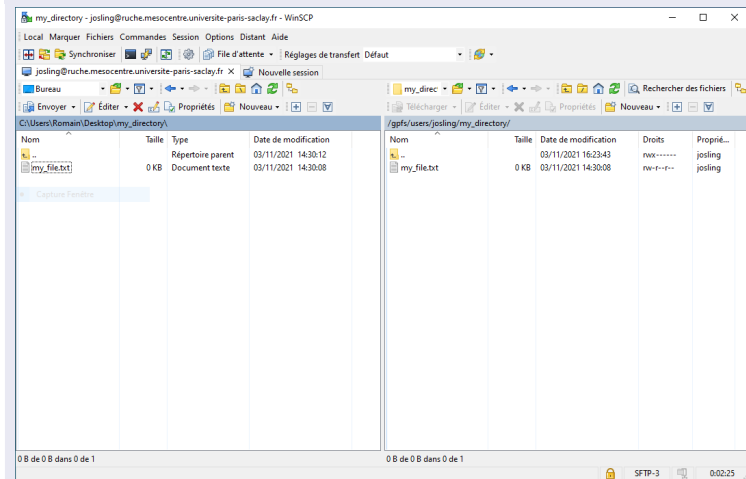


Transfer



SFTP graphical clients: WinSCP

Transfer



- 1 Mésocentre Paris Saclay
- 2 Before computing
 - Connection to Ruche
 - Linux basics commands
 - Text editors
 - Front-end nodes, personal directories, quotas and shared volumes
 - File transfer
 - **The module command**
 - Compilation
 - Python environments
- 3 SLURM Workload manager
- 4 Examples

What is the module command

Several software components (compiler, libraries, software, etc.) are already available on the **Ruche** cluster.

If you do not find what you want, you should to ask the support team to make it available for you.

The module command allows you to customize your environment according to your usage by charging modules. Each loaded module updates your environment variables (such as your *\$PATH* variable) to make a software component reachable by your code.

List all available modules

module avail

```
Terminal - josling@ruche01~  
Fichier  Édition  Affichage  Terminal  Onglets  Aide  
[josling@ruche01 ~]$ module avail  
  
----- /gpfs/softs/modules/modulefiles/compilers -----  
gcc/8.4.0/gcc-4.8.5    gcc/9.2.0/gcc-4.8.5    intel/19.0.3/gcc-4.8.5    intel/20.0.2/gcc-4.8.5    pgi/20.1/gcc-4.8.5  
  
----- /gpfs/softs/modules/modulefiles/debuggers_and_profilers -----  
arm-forge/20.1.3-Redhat-7.0/intel-19.0.3.199    gdb/9.1/gcc-9.2.0    strace/5.2/gcc-9.2.0    valgrind/3.15.0/gcc-9.2.0-intel-mpi  
  
----- /gpfs/softs/modules/modulefiles/languages -----  
anaconda2/2019.10/gcc-9.2.0    openjdk/11.0.2/gcc-9.2.0    python/3.7.6/intel-19.0.3.199  
anaconda3/2020.02/gcc-9.2.0    openjdk/11.0.2/intel-19.0.3.199    r-renv/0.12.2/gcc-9.2.0  
julia/1.4.0/gcc-9.2.0    py-matplotlib/3.1.3/intel-19.0.3.199    r/3.6.2/gcc-9.2.0  
miniconda2/4.7.12.1/intel-19.0.3.199    py-numpy/1.18.1/intel-19.0.3.199    scilab/6.1.0/gcc-9.2.0  
miniconda3/4.7.12.1/intel-19.0.3.199    py-scipy/1.4.1/intel-19.0.3.199  
octave/5.2.0/gcc-9.2.0-intel-mpi    python-nospack/3.6.8/gcc-9.2.0  
  
----- /gpfs/softs/modules/modulefiles/libraries -----  
atlas/3.11.34/gcc-8.4.0    libmesh/1.6.0/gcc-9.2.0-openmpi-debug  
boost/1.69.0/intel-19.0.3.199-intel-mpi-python    libpng/1.6.37/gcc-9.2.0  
boost/1.70.0/intel-19.0.3.199-intel-mpi-python    libxml2/2.9.9/gcc-9.2.0  
boost/1.70.0/intel-19.0.3.199-intel-mpi   metis/5.1.0/intel-19.0.3.199-int32-real64  
boost/1.70.0/intel-19.0.3.199   metis/5.1.0/intel-19.0.3.199-int64-real64  
boost/1.72.0/gcc-9.2.0-openmpi   mmg3d/5.4.3/intel-19.0.3.199  
boost/1.72.0/gcc-9.2.0   motif/2.3.8/gcc-9.2.0  
cantera/2.4.0/intel-19.0.3.199-intel-mpi   mpfr/3.1.6/gcc-9.2.0  
cgal/4.13/intel-19.0.3.199   netcdf-c/4.7.3/intel-19.0.3.199  
cuda/10.0.130/intel-19.0.3.199   netcdf-fortran/4.5.2/intel-19.0.3.199  
cuda/10.1.243/intel-19.0.3.199   netlib-lapack/3.8.0/gcc-9.2.0  
cuda/10.2.89/intel-19.0.3.199   nlopt/2.6.2/gcc-9.2.0  
cuda/9.2.88/intel-19.0.3.199   numactl/2.0.12/gcc-9.2.0  
fftw/3.3.8/gcc-9.2.0-intel-mpi   openblas/0.3.8/gcc-9.2.0  
fftw/3.3.8/intel-19.0.3.199-intel-mpi   openmpi/3.1.5/gcc-9.2.0  
freelut/3.0.0/gcc-9.2.0   openmpi/4.0.2/gcc-9.2.0  
gasnet/2019.3.0/intel-19.0.3.199-intel-mpi   p3dfft3/3.0.0/intel-19.0.3.199-intel-mpi  
gmp/6.1.2/gcc-9.2.0   parmetis/4.0.3/intel-19.0.3.199-intel-mpi-int32-real64  
gmp/6.1.2/intel-19.0.3.199   petsc/3.12.3/gcc-9.2.0-openmpi  
gsl/2.6/gcc-9.2.0   petsc/3.12.3/intel-19.0.3.199-intel-mpi-complex  
hdfs/1.10.6/intel-19.0.3.199-intel-mpi   petsc/3.12.3/intel-19.0.3.199-intel-mpi  
hdfs/1.10.6/intel-19.0.3.199   pnl/1.10.4/intel-19.0.3.199  
hdfs/1.8.21/intel-19.0.3.199-intel-mpi   qhull/2015.2/gcc-9.2.0  
hpx/1.4.0/intel-19.0.3.199-intel-mpi   scotch/6.0.8/gcc-9.2.0-openmpi  
hypre/2.18.2/intel-19.0.3.199-intel-mpi   scotch/6.0.8/intel-19.0.3.199-intel-mpi-esmumps  
intel-mkl/2019.3.199/intel-19.0.3.199-ilp64   scotch/6.0.8/intel-19.0.3.199-intel-mpi  
intel-mkl/2019.3.199/intel-19.0.3.199   suite-sparse/5.7.1/gcc-9.2.0  
intel-mkl/2020.2.254/intel-20.0.2-ilp64   sundials/2.6.2/intel-19.0.3.199-intel-mpi-lapack  
intel-mkl/2020.2.254/intel-20.0.2   sundials/2.6.2/intel-19.0.3.199-intel-mpi
```

module load

```
Terminal - josling@ruche01:~  
Fichier  Edition  Affichage  Terminal  Onglets  Aide  
[josling@ruche01 ~]$ module load intel/20.0.2/gcc-4.8.5  
[josling@ruche01 ~]$ module avail  
  
----- /gpfs/softs/modules/modulefiles/compiler -----  
gcc/8.4.0/gcc-4.8.5      gcc/9.2.0/gcc-4.8.5      intel/19.0.3/gcc-4.8.5      intel/20.0.2/gcc-4.8.5 (L)      pgi/20.1/gcc-4.8.5  
  
----- /gpfs/softs/modules/modulefiles/debuggers_and_profilers -----  
arm-forge/20.1.3-Redhat-7.0/intel-19.0.3.199      gdb/9.1/gcc-9.2.0      strace/5.2/gcc-9.2.0      valgrind/3.15.0/gcc-9.2.0-intel-mpi  
  
----- /gpfs/softs/modules/modulefiles/languages -----  
anaconda2/2019.10/gcc-9.2.0      openjdk/11.0.2/gcc-9.2.0      python/3.7.6/intel-19.0.3.199  
anaconda3/2020.02/gcc-9.2.0      openjdk/11.0.2/intel-19.0.3.199      r-remv/0.12.2/gcc-9.2.0  
julia/1.4.0/gcc-9.2.0      py-matplotlib/3.1.3/intel-19.0.3.199      r/3.6.2/gcc-9.2.0  
miniconda2/4.7.12.1/intel-19.0.3.199      py-numpy/1.18.1/intel-19.0.3.199      scilab/6.1.0/gcc-9.2.0  
miniconda3/4.7.12.1/intel-19.0.3.199      py-scipy/1.4.1/intel-19.0.3.199  
octave/5.2.0/gcc-9.2.0-intel-mpi      python-nospack/3.6.8/gcc-9.2.0  
  
----- /gpfs/softs/modules/modulefiles/libraries -----  
atlas/3.11.34/gcc-8.4.0      libmesh/1.6.0/gcc-9.2.0-openmpi-debug  
boost/1.69.0/intel-19.0.3.199-intel-mpi-python      libpng/1.6.37/gcc-9.2.0  
boost/1.70.0/intel-19.0.3.199-intel-mpi-python      libxml2/2.9.9/gcc-9.2.0  
boost/1.70.0/intel-19.0.3.199-intel-mpi     metis/5.1.0/intel-19.0.3.199-int32-real64  
boost/1.70.0/intel-19.0.3.199     metis/5.1.0/intel-19.0.3.199-int64-real64  
boost/1.72.0/gcc-9.2.0-openmpi      mmg3d/5.4.3/intel-19.0.3.199  
boost/1.72.0/gcc-9.2.0      motif/2.3.8/gcc-9.2.0  
cantera/2.4.0/intel-19.0.3.199-intel-mpi      mpfr/3.1.6/gcc-9.2.0  
cgall/4.13/intel-19.0.3.199      netcdf-c/4.7.3/intel-19.0.3.199  
cuda/10.0.130/intel-19.0.3.199      netcdf-fortran/4.5.2/intel-19.0.3.199  
cuda/10.1.243/intel-19.0.3.199      netlib-lapack/3.8.0/gcc-9.2.0  
cuda/10.2.89/intel-19.0.3.199      nlopt/2.6.2/gcc-9.2.0  
cuda/9.2.88/intel-19.0.3.199      numactl/2.0.12/gcc-9.2.0  
fftw/3.3.8/gcc-9.2.0-intel-mpi      openblas/0.3.8/gcc-9.2.0  
fftw/3.3.8/intel-19.0.3.199-intel-mpi      openmpi/3.1.5/gcc-9.2.0  
freerdp/3.0.0/gcc-9.2.0      openmpi/4.0.2/gcc-9.2.0  
gasnet/2019.3.0/intel-19.0.3.199-intel-mpi      p3dfft/3.0.0/intel-19.0.3.199-intel-mpi  
gmp/6.1.2/gcc-9.2.0      parmetis/4.0.3/intel-19.0.3.199-intel-mpi-int32-real64  
gmp/6.1.2/intel-19.0.3.199      petsc/3.12.3/gcc-9.2.0-openmpi  
gsl/2.6/gcc-9.2.0      petsc/3.12.3/intel-19.0.3.199-intel-mpi-complex  
hdf5/1.10.6/intel-19.0.3.199-intel-mpi      petsc/3.12.3/intel-19.0.3.199-intel-mpi  
hdf5/1.10.6/intel-19.0.3.199      pnl/1.10.4/intel-19.0.3.199  
hdf5/1.8.21/intel-19.0.3.199-intel-mpi      qhull/2015.2/gcc-9.2.0  
hpx/1.4.0/intel-19.0.3.199-intel-mpi      scotch/6.0.8/gcc-9.2.0-openmpi  
hypre/2.18.2/intel-19.0.3.199-intel-mpi      scotch/6.0.8/intel-19.0.3.199-intel-mpi-esmumps  
intel-mkl/2019.3.199/intel-19.0.3.199-ilp64      scotch/6.0.8/intel-19.0.3.199-intel-mpi  
intel-mkl/2019.3.199/intel-19.0.3.199      suite-sparse/5.7.1/gcc-9.2.0  
intel-mkl/2020.2.254/intel-20.0.2-ilp64      sundials/2.6.2/intel-19.0.3.199-intel-mpi-lapack
```

Show all loaded modules

module list

Terminal - josling@ruche01:~

```
Fichier  Édition  Affichage  Terminal  Onglets  Aide
[josling@ruche01 ~]$ module load intel/20.0.2/gcc-4.8.5
[josling@ruche01 ~]$ module list
```

```
Currently Loaded Modules:
 1) intel/20.0.2/gcc-4.8.5
```

```
[josling@ruche01 ~]$
```

Unload modules

module unload

Terminal - josling@ruche01:~

```
Fichier  Édition  Affichage  Terminal  Onglets  Aide
[josling@ruche01 ~]$ module load intel/20.0.2/gcc-4.8.5
[josling@ruche01 ~]$ module list
```

Currently Loaded Modules:

1) intel/20.0.2/gcc-4.8.5

```
[josling@ruche01 ~]$ module unload intel/20.0.2/gcc-4.8.5
```

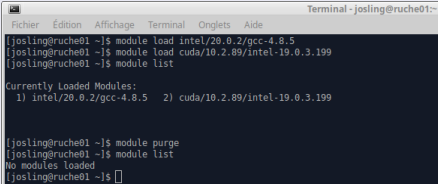
```
[josling@ruche01 ~]$ module list
```

No modules loaded

```
[josling@ruche01 ~]$
```

Unload modules

module purge

A terminal window titled "Terminal - josling@ruche01:~" with a menu bar containing "Fichier", "Édition", "Affichage", "Terminal", "Onglets", and "Aide". The terminal shows a sequence of commands: "module load intel/20.0.2/gcc-4.8.5", "module load cuda/10.2.89/intel-19.0.3.199", and "module list". The output of "module list" shows two loaded modules. Then, the "module purge" command is entered, followed by "module list", which outputs "No modules loaded".

```
Terminal - josling@ruche01:~  
Fichier  Édition  Affichage  Terminal  Onglets  Aide  
[josling@ruche01 ~]$ module load intel/20.0.2/gcc-4.8.5  
[josling@ruche01 ~]$ module load cuda/10.2.89/intel-19.0.3.199  
[josling@ruche01 ~]$ module list  
  
Currently Loaded Modules:  
  1) intel/20.0.2/gcc-4.8.5  2) cuda/10.2.89/intel-19.0.3.199  
  
[josling@ruche01 ~]$ module purge  
[josling@ruche01 ~]$ module list  
No modules loaded  
[josling@ruche01 ~]$
```

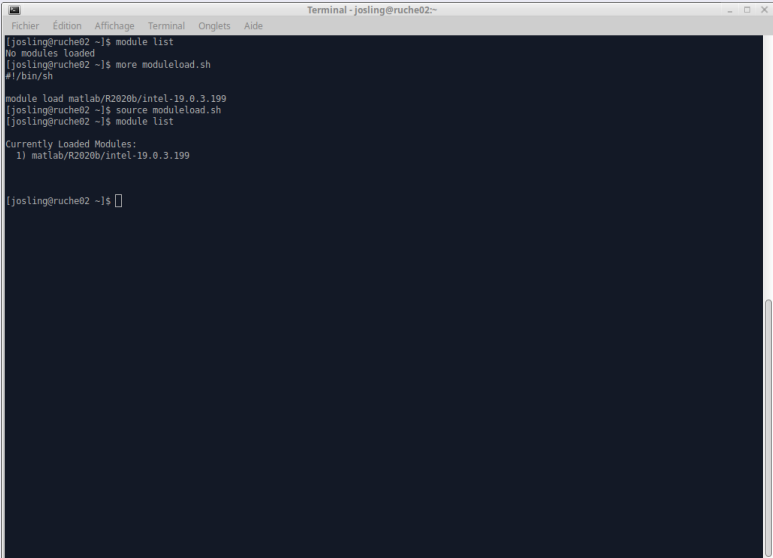
**DO NOT EDIT *.bashrc*, *.bash_login*
NOR *.bash_profile***

What to do then

- Create a file containing the module load commands and source it.
- Use the collection mechanism of module

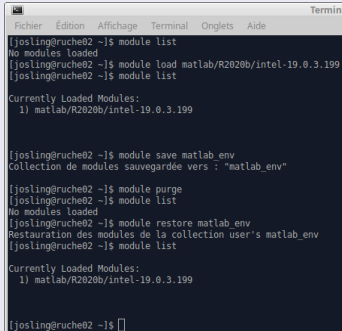
Frequently load modules

File sourcing

A terminal window titled "Terminal - josling@ruche02:~" with a menu bar containing "Fichier", "Édition", "Affichage", "Terminal", "Onglets", and "Aide". The terminal shows a sequence of commands and their outputs: "module list" returns "No modules loaded"; "more moduleload.sh" opens a shell; "module load matlab/R2020b/intel-19.0.3.199" loads the module; "source moduleload.sh" sources the script; and a second "module list" shows "Currently Loaded Modules: 1) matlab/R2020b/intel-19.0.3.199".

```
Terminal - josling@ruche02:~  
Fichier  Édition  Affichage  Terminal  Onglets  Aide  
josling@ruche02 ~]$ module list  
No modules loaded  
josling@ruche02 ~]$ more moduleload.sh  
#!/bin/sh  
  
module load matlab/R2020b/intel-19.0.3.199  
josling@ruche02 ~]$ source moduleload.sh  
josling@ruche02 ~]$ module list  
  
Currently Loaded Modules:  
1) matlab/R2020b/intel-19.0.3.199  
  
josling@ruche02 ~]$
```


Collection mechanism



```
Terminal - josling@ruche02:~  
Fichier  Édition  Affichage  Terminal  Onglets  Aide  
[josling@ruche02 ~]$ module list  
No modules loaded  
[josling@ruche02 ~]$ module load matlab/R2020b/intel-19.0.3.199  
[josling@ruche02 ~]$ module list  
  
Currently Loaded Modules:  
  1) matlab/R2020b/intel-19.0.3.199  
  
[josling@ruche02 ~]$ module save matlab_env  
Collection de modules sauvegardée vers : "matlab_env"  
  
[josling@ruche02 ~]$ module purge  
[josling@ruche02 ~]$ module list  
No modules loaded  
[josling@ruche02 ~]$ module restore matlab_env  
Restauration des modules de la collection user's matlab_env  
[josling@ruche02 ~]$ module list  
  
Currently Loaded Modules:  
  1) matlab/R2020b/intel-19.0.3.199  
  
[josling@ruche02 ~]$
```

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Sequential code

```
> module load intel/19.0.3/gcc-4.8.5
> ifort -o hello_fortran hello_seq.f90
> icc -o hello_c hello_seq.c
> icpc -o hello_cpp hello_seq.cpp
```

OpenMP code

```
> module load intel/19.0.3/gcc-4.8.5
> ifort -qopenmp -o hello_omp_fortran hello_omp.f90
> icc -qopenmp -o hello_omp_c hello_omp.c
> icpc -qopenmp -o hello_omp_cpp hello_omp.cpp
```

MPI code

```
> module load intel/19.0.3/gcc-4.8.5
> module load intel-mpi/2019.3.199/intel-19.0.3.199
> mpiifort -o hello_mpi_fortran hello_mpi.f90
> mpiicc -o hello_mpi_c hello_mpi.c
> mpiicpc -o hello_mpi_cpp hello_mpi.cpp
```

Sequential code

```
> module load gcc/9.2.0/gcc-4.8.5
> gfortran -o hello_fortran hello_seq.f90
> gcc -o hello_c hello_seq.c
> g++ -o hello_cpp hello_seq.cpp
```

OpenMP code

```
> module load gcc/9.2.0/gcc-4.8.5
> gfortran -fopenmp -o hello_omp_fortran hello_omp.f90
> gcc -fopenmp -o hello_omp_c hello_omp.c
> g++ -fopenmp -o hello_omp_cpp hello_omp.cpp
```

MPI code

```
> module load gcc/9.2.0/gcc-4.8.5
> module load openmpi/4.0.2/gcc-9.2.0
> mpifort -o hello_mpi_fortran hello_mpi.f90
> mpicc -o hello_mpi_c hello_mpi.c
> mpic++ -o hello_mpi_cpp hello_mpi.cpp
```

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Conda modules

```
Terminal - josling@ruche01:~  
Fichier Édition Affichage Terminal Onglets Aide  
[josling@ruche01 ~]$ module avail anaconda miniconda  
----- /gpfs/softs/modules/modulefiles/languages -----  
anaconda2/2019.10/gcc-9.2.0  anaconda3/2020.02/gcc-9.2.0  miniconda2/4.7.12.1/intel-19.0.3.199  miniconda3/4.7.12.1/intel-19.0.3.199  
  
Utilisez "module spider" pour trouver tous les modules possibles.  
Utilisez "module keyword key1 key2 ..." pour chercher tous les modules possibles qui correspondent à l'une des clés (key1, key2).  
  
[josling@ruche01 ~]$ module load anaconda3/2020.02/gcc-9.2.0  
[josling@ruche01 ~]$ conda list  
# packages in environment at /gpfs/softs/languages/anaconda3/2020.02:  
#  
# Name                               Version      Build      Channel  
ipyw_jlab_nb_ext_conf               0.1.0        py37_0  
libgcc_mutex                         0.1          main  
alabaster                           0.7.12       py37_0  
anaconda                            2020.02      py37_0  
anaconda-client                     1.7.2        py37_0  
anaconda-navigator                  1.9.12       py37_0  
anaconda-project                    0.8.4        py_0  
argh                                 0.26.2       py37_0  
asn1crypto                          1.3.0        py37_0  
astroid                             2.3.3        py37_0  
astropy                             4.0          py37h7b6447c_0  
atomicwrites                        1.3.0        py37_1  
attrs                               19.3.0       py_0  
autopep8                            1.4.4        py_0  
babel                               2.8.0        py_0  
backcall                            0.1.0        py37_0  
backports                           1.0          py_2  
backports.functools_lru_cache 1.6.1        py_0  
backports.shutil_get_terminal_size 1.0.0        py37_2  
backports.tempfile                  1.0          py_1  
backports.weakref                    1.0.post1    py_1  
beautifulsoup4                      4.8.2        py37_0  
bitarray                            1.2.1        py37h7b6447c_0  
bkcharts                            0.2          py37_0  
blas                                 1.0          mkl  
bleach                              3.1.0        py37_0  
blosc                               1.16.3       hd408876_0  
bokeh                                1.4.0        py37_0  
boto                                 2.49.0       py37_0  
bottleneck                          1.3.2        py37heb32a55_0  
bzip2                               1.0.8        h7b6447c_0  
ca-certificates                     2020.1.1     0
```

Environment creation

```
Terminal - josling@ruche01:~
Fichier  Édition  Affichage  Terminal  Onglets  Aide
[josling@ruche01 ~]$ module load anaconda3/2020.02/gcc-9.2.0
[josling@ruche01 ~]$ conda create -n my_env
Collecting package metadata (current_repodata.json): done
Solving environment: done

==> WARNING: A newer version of conda exists. <==
  current version: 4.8.2
  latest version: 4.9.2

Please update conda by running

  $ conda update -n base -c defaults conda

## Package Plan ##

  environment location: /gpfs/users/josling/.conda/envs/my_env

Proceed ([y]/n)? y
Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#   $ conda activate my_env
#
# To deactivate an active environment, use
#
#   $ conda deactivate
#
[josling@ruche01 ~]$ source activate my_env
(my_env) [josling@ruche01 ~]$
```

Do not use *conda activate my_env*

followed by *conda init bash* to activate an environment: this would modify your */gpfs/users/username/.bashrc* file and may result in a conflict with other environments and conda updates

Use *source activate my_env* instead

Installing packages

```
Terminal - josling@ruche01:~  
Fichier  Édition  Affichage  Terminal  Onglets  Aide  
[josling@ruche01 ~]$ source activate my_env  
(my env) [josling@ruche01 ~]$ conda install mpi4py  
Collecting package metadata (current_repodata.json): done  
Solving environment: done  
  
==> WARNING: A newer version of conda exists. <==  
current version: 4.8.2  
latest version: 4.9.2  
  
Please update conda by running  
  
$ conda update -n base -c defaults conda  
  
## Package Plan ##  
  
environment location: /gpfs/users/josling/.conda/envs/my_env  
  
added / updated specs:  
- mpi4py  
  
The following packages will be downloaded:  
  
package | build  
-----|-----  
mpi-1.0 | mpich | 13 KB  
mpi4py-3.0.3 | py37hf046dal_1 | 536 KB  
mpich-3.3.2 | hc856adb_0 | 3.8 MB  
-----|-----  
Total: | 4.4 MB  
  
The following NEW packages will be INSTALLED:  
  
libgcc_mutex | pkgs/main/linux-64::libgcc_mutex-0.1-main  
ca-certificates | pkgs/main/linux-64::ca-certificates-2021.1.19-h06a4308_0  
certifi | pkgs/main/linux-64::certifi-2020.12.5-py37h06a4308_0  
ld_impl_linux-64 | pkgs/main/linux-64::ld_impl_linux-64-2.33.1-h53a641e_7  
libedit | pkgs/main/linux-64::libedit-3.1.20191231-h14c3975_1  
libffi | pkgs/main/linux-64::libffi-3.3-he6710b0_2  
libgcc-ng | pkgs/main/linux-64::libgcc-ng-9.1.0-hdf63c60_0  
libgfortran-ng | pkgs/main/linux-64::libgfortran-ng-7.3.0-hdf63c60_0  
libstdcxx-ng | pkgs/main/linux-64::libstdcxx-ng-9.1.0-hdf63c60_0  
mpi | pkgs/main/linux-64::mpi-1.0-mpich
```

Installing packages

```
Terminal - josling@ruche01:~  
Fichier  Édition  Affichage  Terminal  Onglets  Aide  
mpi          pkgs/main/linux-64::mpi-1.0-mpich  
mpi4py       pkgs/main/linux-64::mpi4py-3.0.3-py37hf046da1_1  
mpich        pkgs/main/linux-64::mpich-3.3.2-hc856adb_0  
ncurses      pkgs/main/linux-64::ncurses-6.2-he6710b0_1  
openssl      pkgs/main/linux-64::openssl-1.1.1i-h27cfd23_0  
pip          pkgs/main/linux-64::pip-20.3.3-py37h06a4308_0  
python       pkgs/main/linux-64::python-3.7.9-h7579374_0  
readline     pkgs/main/linux-64::readline-8.1-h27cfd23_0  
setuptools   pkgs/main/linux-64::setuptools-52.0.0-py37h06a4308_0  
sqlite       pkgs/main/linux-64::sqlite-3.33.0-h62c20be_0  
tk           pkgs/main/linux-64::tk-8.6.10-hbc83047_0  
wheel        pkgs/main/noarch::wheel-0.36.2-pyhd3eb1b0_0  
xz           pkgs/main/linux-64::xz-5.2.5-h7b6447c_0  
zlib         pkgs/main/linux-64::zlib-1.2.11-h7b6447c_3  
  
Proceed ([y]/n)? y  
  
Downloading and Extracting Packages  
mpi-1.0                | 13 KB | ##### | 100%  
mpi4py-3.0.3           | 536 KB | ##### | 100%  
mpich-3.3.2            | 3.8 MB | ##### | 100%  
Preparing transaction: done  
Verifying transaction: done  
Executing transaction: done  
(my env) [josling@ruche01 ~]$ conda list  
# packages in environment at /gpfs/users/josling/.conda/envs/my_env:  
#  
# Name                   Version           Build    Channel  
_libgcc_mutex            0.1               main  
ca-certificates          2021.1.19         h06a4308_0  
certifi                  2020.12.5         py37h06a4308_0  
ld_impl_linux-64        2.33.1            h53a641e_7  
libedit                  3.1.20191231      h14c3975_1  
libffi                   3.3               he6710b0_2  
libgcc-ng                9.1.0             hdf63c60_0  
libgfortran-ng           7.3.0             hdf63c60_0  
libstdcxx-ng             9.1.0             hdf63c60_0  
mpi                       1.0               mpich  
mpi4py                   3.0.3             py37hf046da1_1  
mpich                    3.3.2             hc856adb_0  
ncurses                   6.2               he6710b0_1  
openssl                   1.1.1i            h27cfd23_0  
pip                       20.3.3            py37h06a4308_0  
python                    3.7.9             h7579374_0
```

Anaconda installs all the package into the directory *\$HOME/.conda*.

These files therefore restrain your *\$HOME* quota (50 GB).

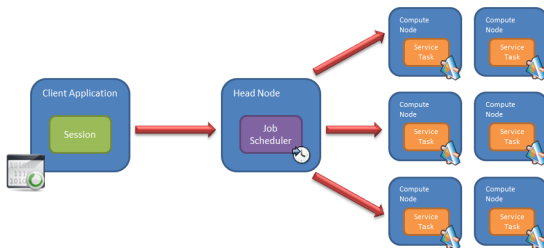
Since large conda environments (like IA frameworks pytorch/keras/tensorflow) can take several GB of data, you may need to move this directory to your workdir space and create a symlink.

```
[username@ruche ~]$ mv $HOME/.conda $WORKDIR/.conda  
[username@ruche ~]$ ln -s $WORKDIR/.conda ~/.conda
```

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Resource management

- More than 300 users from 54 different labs perform computations on **Ruche** .



- The scheduler, a program responsible for the distribution of jobs according to the available resources. According to the policy defined on the machine, it chooses when to execute the jobs and with which resources.
- The user provides the scheduler a description of his job (script, necessary resources, maximum execution time, other constraints) with a specific syntax.
- Large-scale computing centers all work with a scheduler.

Configuration of the Resource Manager

Scheduler policy

- Key point of the good use of the machine to respect the objectives of **Ruche**
- Evolves based on experience feedback

Parametrization characteristics

- Limiting jobs execution time in order to:
 - enable job rotation
 - impel users to implement a restart procedure in their codes
- Dedicating a large number of nodes for parallel job in order to:
 - favor jobs that require a large number of cores
 - impel users to use parallelization techniques
- Dedicating some nodes for daytime tests in order to:
 - reduce the waiting time for jobs that require less than 1 hour
 - support code development
- Enabling fairshare
 - priority to users/labs with the least calculation for the last 30 days
 - score for each partner, according to the partnership agreement
 - sub-score by laboratory, decided by the partners

Energy sobriety policy

To participate in the energy sobriety effort of our establishments, 1/3 of **Ruche** resources has been turned off:

- 72/216 cpu nodes
- 4/14 large memory nodes
- 10/20 fusion large memory nodes
- 4/4 P100 gpu nodes
- 4/10 V100 gpu nodes
- 2/9 A100 gpu nodes
- 1/3 visualization nodes

Slurm partitions - CPU

Queue name	Pool	Max. memory per user**	Max CPU per user**	Max Walltime	Node equivalent
cpu_short*	cpu	4.5 TB	1000 cpus	01:00:00	25
cpu_med	cpu	4.5 TB	1000 cpus	04:00:00	25
cpu_long	cpu	720 GB	160 cpus	168:00:00	4
cpu_prod****	cpu	9 TB	2000 cpus	06:00:00	50
cpu_scale*/****	cpu	18 TB	4000 cpus	01:00:00	100
mem_short*	mem	1.5 TB	80 cpus	01:00:00	1
mem	mem	3 TB	160 cpus	72:00:00	2
fusion_shm***	fusion	1.5 TB	160 cpus	168:00:00	4
fusion_shm_big*/***	fusion	1.5 TB	320 cpus	168:00:00	8

* : 1 job/user

** : This rule applies on running jobs of a user on the specified partition

*** : This partition is formed with heterogenous compute nodes

**** : Active during the week from 8pm to 8am and all day long in weekends

Slurm partitions - GPU

Queue name	Pool	Nb of units per node	Max. memory per node	Max GPUs per user**	Max Walltime
gpu_test*	V100	40 cpus 4 gpus	756 GB	8 gpus	01:00:00
gpu	V100	40 cpus 4 gpus	756 GB	8 gpus	24:00:00
gpua100	A100	32 cpus 4 gpus	1 TB	4 gpus	24:00:00
gpup100	P100	24 cpus 2 gpus	180 GB	2 gpus	168:00:00

* : 1 job/user

** : This rule applies on running jobs of a user on the specified partition.

Max CPU per GPU reserved for a job = Nb cpus per node / Nb gpus per node

Slurm directives: allocate resources (I)

Number of nodes

```
#SBATCH --nodes=<nnodes>
```

Number of tasks (MPI processes)

```
#SBATCH --ntasks=<ntasks>
```

Number of tasks (MPI processes) per node

```
#SBATCH --ntasks-per-node=<ntpn>
```

Number of threads per task (Ex: Openmp threads per MPI process)

```
#SBATCH --cpus-per-task=<ntpt>
```

Slurm directives: allocate resources (II)

Number of gpus

```
#SBATCH --gres=gpu:<ngpus>
```

Memory per node

```
#SBATCH --mem=<memory>
```

Default memory is 4 GB per core.

Walltime

```
#SBATCH --time=<hh:mm:ss>
```

If your job is still running after the walltime duration, your job will be killed.

Partition

```
#SBATCH --partition=<PartitionName>
```

Slurm directives: miscellaneous (I)

Define the job's name

```
#SBATCH --job-name=<JobName>
```

Define the standard output (stdout) for your job

```
#SBATCH --output=<FileName>
```

Define the error output (stderr) for your job

```
#SBATCH --error=<FileName>
```

By default both standard output and standard error are directed to the same file.

Reserve tempory space on the SSD (only for mem and gpu nodes)

```
#SBATCH --tmp=<size[units]>
```

By default, tmp will be 100M.

Set an email address

```
#SBATCH --mail-user=<email-address>
```

Be notified by mail when a step has been reached

```
#SBATCH --mail-type=<args>
```

Arguments for *--mail-type* option are :

- BEGIN : send an email when the job starts
- END : send an email when the job stops
- FAIL : send an email if the job fails
- ALL : equivalent to BEGIN, END, FAIL

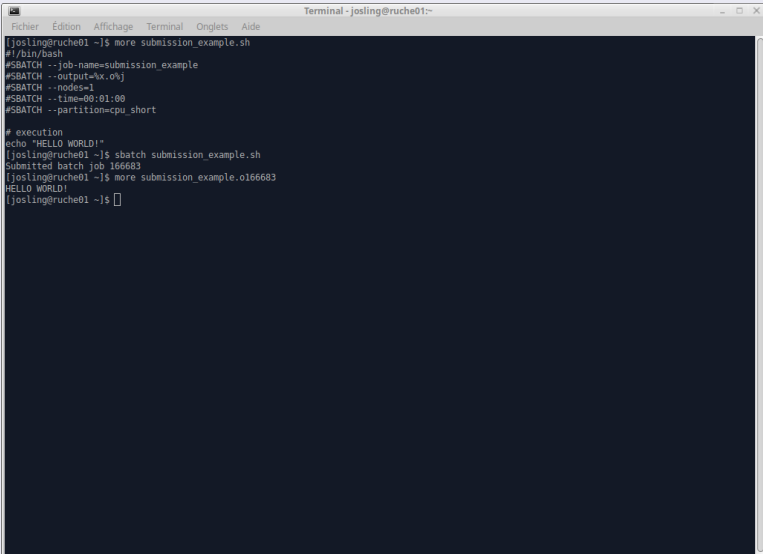
Export user environment variables

- By default all user environment variables will be loaded:
`#SBATCH --export=ALL`
- Export no environment variables (recommended):
`#SBATCH --export=NONE`
- To select explicitly export variables:
`#SBATCH --export=VAR1,VAR2`

Propagate resources limits

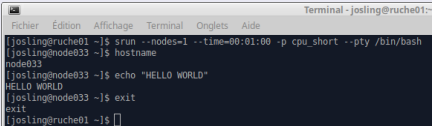
- By default all resources limits are propagated:
`#SBATCH --propagate=ALL`
- To avoid the propagation of resources limits (recommended):
`#SBATCH --propagate=NONE`

Submit a slurm script



```
Terminal - josling@ruche01:~  
Fichier  Édition  Affichage  Terminal  Onglets  Aide  
[josling@ruche01 ~]$ more submission_example.sh  
#!/bin/bash  
#SBATCH --job-name=submission_example  
#SBATCH --output=%x.%j  
#SBATCH --nodes=1  
#SBATCH --time=00:01:00  
#SBATCH --partition=cpu_short  
  
# execution  
echo "HELLO WORLD!"  
[josling@ruche01 ~]$ sbatch submission_example.sh  
Submitted batch job 166683  
[josling@ruche01 ~]$ more submission_example.o166683  
HELLO WORLD!  
[josling@ruche01 ~]$
```

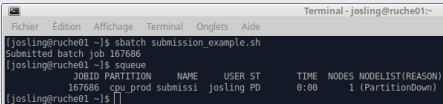
Request an interactive session



A terminal window titled "Terminal - josling@ruche01:-" with a menu bar containing "Fichier", "Édition", "Affichage", "Terminal", "Onglets", and "Aide". The terminal shows the following sequence of commands and output:

```
[josling@ruche01 ~]$ srun --nodes=1 --time=00:01:00 -p cpu_short --pty /bin/bash
[josling@node033 ~]$ hostname
node033
[josling@node033 ~]$ echo "HELLO WORLD"
HELLO WORLD
[josling@node033 ~]$ exit
exit
[josling@ruche01 ~]$
```


Monitor jobs: *squeue*

A terminal window titled "Terminal - josling@ruche01~" with a menu bar (Fichier, Édition, Affichage, Terminal, Onglets, Aide). The terminal shows the execution of 'sbatch submission_example.sh', which submits a batch job 167686. Then, 'squeue' is run, displaying a table of jobs. The table has columns: JOBID, PARTITION, NAME, USER, ST, TIME, NODES, and NODELIST(REASON). One job is listed: 167686, cpu_prod, submissi, josling, PD, 0:00, 1, (PartitionDown).

```
[josling@ruche01 ~]$ sbatch submission_example.sh
Submitted batch job 167686
[josling@ruche01 ~]$ squeue
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
167686	cpu_prod	submissi	josling	PD	0:00	1	(PartitionDown)

```
[josling@ruche01 ~]$
```

Monitor jobs: *squeue*

```
Terminal - josling@ruche01:~  
Fichier  Édition  Affichage  Terminal  Onglets  Aide  
[josling@ruche01 ~]$ sbatch submission_example.sh  
Submitted batch job 167686  
[josling@ruche01 ~]$ squeue  
JOBID PARTITION   NAME     USER ST       TIME  NODES NODELIST(REASON)  
167686 cpu_prod submissi josling PD        0:00      1 (PartitionDown)  
[josling@ruche01 ~]$ squeue  
JOBID PARTITION   NAME     USER ST       TIME  NODES NODELIST(REASON)  
167686 cpu_prod submissi josling R        0:21      1 node019  
[josling@ruche01 ~]$
```

Cancel a job: *scancel*

```
Terminal - josling@ruche01:~  
Fichier  Édition  Affichage  Terminal  Onglets  Aide  
[josling@ruche01 ~]$ sbatch submission_example.sh  
Submitted batch job 167686  
[josling@ruche01 ~]$ squeue  
      JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)  
      167686 cpu_prod  submissi  josling PD        0:00      1 (PartitionDown)  
[josling@ruche01 ~]$ squeue  
      JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)  
      167686 cpu_prod  submissi  josling R        0:21      1 node019  
[josling@ruche01 ~]$ scancel 167686  
[josling@ruche01 ~]$ squeue  
      JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)  
[josling@ruche01 ~]$
```

Job arrays: *sbatch* option `--array` (or `-a`)

Jobs must be independent of each other.

Submit a job array with index values between 0 and 31

```
[username@ruche01 ~]$ sbatch --array=0-31 job
```

Submit a job array with index values of 1, 3, 5 and 7

```
[username@ruche01 ~]$ sbatch --array=1,3,5,7 job
```

Submit a job array with index values between 1 and 7 with a step size of 2 (i.e. 1, 3, 5 and 7)

```
[username@ruche01 ~]$ sbatch --array=1-7:2 job
```

`SLURM_ARRAY_TASK_ID` is the job's index value in the array.

Chain jobs: *sbatch* option *--dependency*

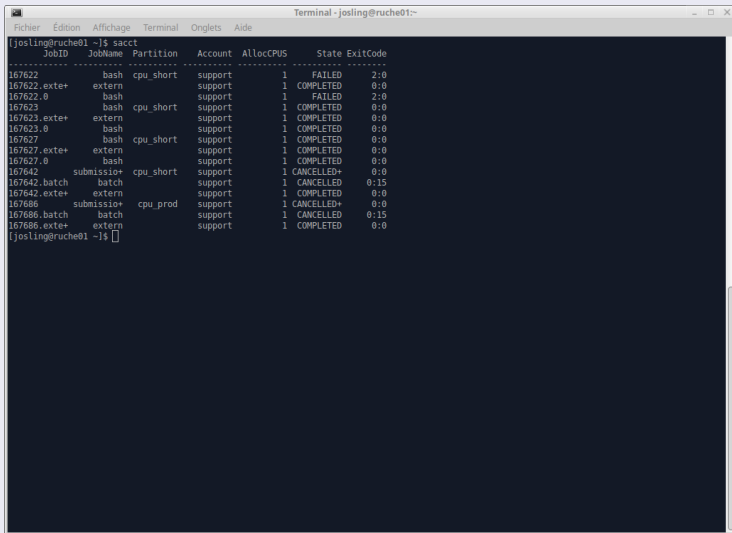
Common chaining rules

- after: = job can start once job has started execution
- afterany: = job can start once job has terminated
- afterok: = job can start once job has terminated successfully
- afternotok: = job can start once job has terminated upon failure
- singleton = job can start once any previous job with identical name and user has terminated

Example: Start of job after another job has successfully finished

```
[username@ruche01 ~]$ sbatch slurm_script1.sh
Submitted batch job 74698
[username@ruche01 ~]$ sbatch --dependency=afterok:74698 slurm_script2.sh
Submitted batch job 74699
```

Get info on your finished jobs: *sacct*



```
Terminal - josling@ruche01:~
[josling@ruche01 ~]$ sacct
JobID JobName Partition Account AllocCPUS State ExitCode
-----
167622 bash cpu_short support 1 FAILED 2:0
167622.exte+ extern support 1 COMPLETED 0:0
167622.0 bash support 1 FAILED 2:0
167623 bash cpu_short support 1 COMPLETED 0:0
167623.exte+ extern support 1 COMPLETED 0:0
167623.0 bash support 1 COMPLETED 0:0
167627 bash cpu_short support 1 COMPLETED 0:0
167627.exte+ extern support 1 COMPLETED 0:0
167627.0 bash support 1 COMPLETED 0:0
167642 submission+ cpu_short support 1 CANCELLED+ 0:0
167642.batch batch support 1 CANCELLED 0:15
167642.exte+ extern support 1 COMPLETED 0:0
167686 submission+ cpu_prod support 1 CANCELLED+ 0:0
167686.batch batch support 1 CANCELLED 0:15
167686.exte+ extern support 1 COMPLETED 0:0
[josling@ruche01 ~]$
```

Note that it will only cover the last 30 days.

Get details on one finished job: *seff*

Terminal - josling@ruche01:~

```
Fichier  Édition  Affichage  Terminal  Onglets  Aide
[josling@ruche01 ~]$ seff 167622
Job ID: 167622
Cluster: ruche
User/Group: josling/mics
State: FAILED (exit code 130)
Cores: 1
CPU Utilized: 00:00:00
CPU Efficiency: 0.00% of 00:00:04 core-walltime
Job Wall-clock time: 00:00:04
Memory Utilized: 1.78 MB
Memory Efficiency: 0.04% of 4.00 GB
[josling@ruche01 ~]$
```

About priorities

Three factors are used to calculate job priority:

Age

The longer a job waits for resources to become available, the higher the priority it becomes. Maximum reached in 3 days.

Fairshare

- The machine is divided into 29% for CS, ENS and UPSay and 13% for MDLS
- Within partners, a weight can be assigned by laboratory (Ex: depending on participation for CS)
- The more a user, his laboratory and/or the partner on which he depends, has calculated lately, the less his jobs have priority

Size

A job requiring a lot of resources has a little more priority to compensate for its difficulty in placing itself on resources not used by smaller jobs.

These factors apply to everyone at all times except for administration. There are no tricks to avoid them nor privileges.

Energy sobriety policy

To participate as well as possible in the energetic performance of **Ruche**, we ask you to be vigilant when running your calculations:

- properly target the resources you request
- optimize your different codes
- priority use recent resources, more energy efficient (gpu A100 rather than V100, cpu resources rather than fusion_shm)

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- ▶ Job arrays

- ▶ Bash

- ▶ Sequential

- ▶ MPI

- ▶ OpenMP

- ▶ MPI + OpenMP

- ▶ Cuda

- ▶ Matlab

- ▶ Julia

- ▶ Python mpi4py

- ▶ Python numpy

- ▶ Python Keras

- ▶ Python Pytorch

- Etc.

See https://gitlab-research.centralesupelec.fr/mesocentre-public/ruche_examples for more

Resources

1 CPU (1 task) on 1 node for 20 min on cpu_short

Output

node100

mer. nov. 23 10:58:33 CET 2022

Accounting

Job ID: 2731304

Cluster: ruche

User/Group: josling/mics

State: COMPLETED (exit code 0)

Cores: 1

CPU Utilized: 00:00:00

CPU Efficiency: 0.00% of 00:00:20 core-walltime

Job Wall-clock time: 00:00:20

Memory Utilized: 1.73 MB

Memory Efficiency: 0.04% of 4.00 GB

Job arrays

Resources

1 CPU (1 task) on 1 node for 20 min on cpu_short | x 10

Output

Hello from array 1 of 10
node141
mer. nov. 23 11:10:20 CET 2022 | x 10

Accounting

Job ID: 2731312
Array Job ID: 2731312_10
Cluster: ruche
User/Group: josling/mics
State: COMPLETED (exit code 0)
Cores: 1
CPU Utilized: 00:00:00
CPU Efficiency: 0.00% of 00:00:21 core-walltime
Job Wall-clock time: 00:00:21
Memory Utilized: 1.74 MB
Memory Efficiency: 0.04% of 4.00 GB | x 10

Resources

2 CPUs (2 threads on 1 task) on 1 node for 20 min on cpu_short

Output

LASTN =

2

Elapsed time is 51.217917 seconds.

Accounting

Job ID: 2732120

Cluster: ruche

User/Group: josling/mics

State: COMPLETED (exit code 0)

Nodes: 1

Cores per node: 2

CPU Utilized: 00:01:56

CPU Efficiency: 48.74% of 00:03:58 core-walltime

Job Wall-clock time: 00:01:59

Memory Utilized: 6.59 GB

Memory Efficiency: 82.34% of 8.00 GB

Resources

40 CPUs (10 threads on 4 tasks) on 2 nodes for 20 min on `cpu_short`

Output

```
Hello from thread 3 out of 10 from process 2 out of 4
Hello from thread 5 out of 10 from process 0 out of 4
Hello from thread 9 out of 10 from process 1 out of 4
Hello from thread 4 out of 10 from process 3 out of 4
Hello from thread 0 out of 10 from process 2 out of 4
...
```

Accounting

```
Job ID: 2732135
Cluster: ruche
User/Group: josling/mics
State: COMPLETED (exit code 0)
Nodes: 2
Cores per node: 20
CPU Utilized: 00:00:05
CPU Efficiency: 3.12% of 00:02:40 core-walltime
Job Wall-clock time: 00:00:04
Memory Utilized: 222.72 MB (estimated maximum)
Memory Efficiency: 0.14% of 160.00 GB (4.00 GB/core)
```

Resources

1 CPU (1 task) and 1 GPU on 1 node for 20 min on gpu_test

Output

Hello Hello

Accounting

Job ID: 2732153

Cluster: ruche

User/Group: josling/mics

State: COMPLETED (exit code 0)

Cores: 1

CPU Utilized: 00:00:01

CPU Efficiency: 12.50% of 00:00:08 core-walltime

Job Wall-clock time: 00:00:08

Memory Utilized: 1.75 MB

Memory Efficiency: 0.04% of 4.00 GB

Resources

4 CPU (4 task) on 1 node for 30 min on `cpu_short`

Output

```
Files already downloaded and verified
Files already downloaded and verified
[1, 2000] loss: 2.232
[...]
[2, 12000] loss: 1.323
Finished Training
Accuracy of the network on the 10000 test images: 53 %
Accuracy of plane : 46 %
[...]
Accuracy of truck : 60 %

real 3m4.249s
user 9m44.439s
sys 0m14.305s
```

Resources

4 CPU (4 task) on 1 node for 30 min on `cpu_short`

Accounting

Job ID: 2757483

Cluster: `ruce`

User/Group: `josling/mics`

State: `COMPLETED` (exit code 0)

Nodes: 1

Cores per node: 4

CPU Utilized: 00:10:00

CPU Efficiency: 78.12% of 00:12:48 core-walltime

Job Wall-clock time: 00:03:12

Memory Utilized: 1.36 GB

Memory Efficiency: 8.51% of 16.00 GB

Resources

4 CPU (4 task) and 1 GPU on 1 node for 30 min on gpu_test

Output

```
cuda:0
Files already downloaded and verified
Files already downloaded and verified
[1, 2000] loss: 2.163
[...]
[2, 12000] loss: 1.261
Finished Training
Accuracy of the network on the 10000 test images: 53 %
Accuracy of plane : 46 %
[...]
Accuracy of truck : 75 %

real 2m56.367s
user 3m33.647s
sys 0m24.025s
```

Resources

4 CPU (4 task) and 1 GPU on 1 node for 30 min on `gpu_test`

Accounting

Job ID: 2756762

Cluster: `ruce`

User/Group: `josling/mics`

State: `COMPLETED` (exit code 0)

Nodes: 1

Cores per node: 4

CPU Utilized: 00:03:59

CPU Efficiency: 32.12% of 00:12:24 core-walltime

Job Wall-clock time: 00:03:06

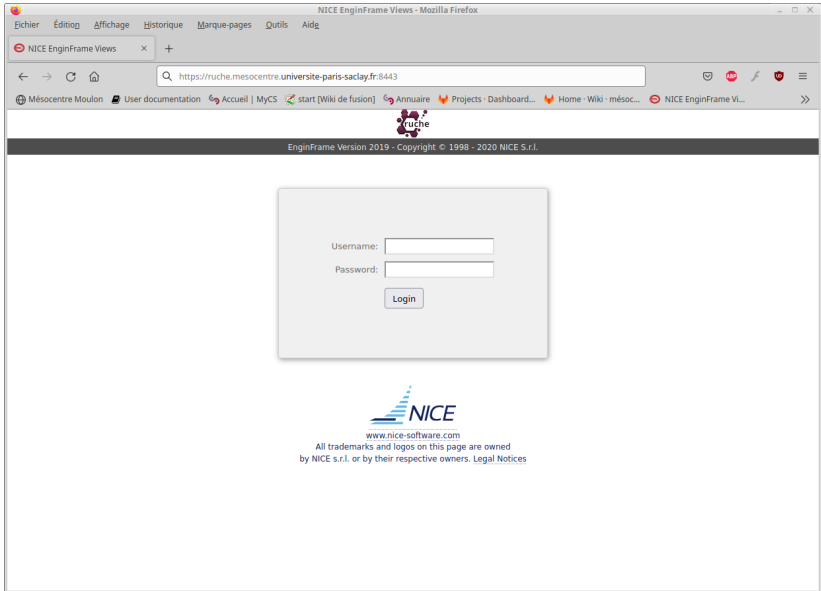
Memory Utilized: 15.60 GB

Memory Efficiency: 97.51% of 16.00 GB

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Browser	Nice DCV Desktop Client
In a web browser, go to https://ruche.mesocentre.universite-paris-saclay.fr:8443	
Log into the web interface with your Ruche account and your password	
In the left-side menu, click on "Linux Desktop"	In the top right corner in Settings, in Nice DCV (since 2017), Select "Desktop Client" the type of NICE DCV viewer you want to use to connect to a remote session.
Define the walltime and the resources you need	
Hit the "Launch Session" button	
Your session is created A session representation will appear on the web interface Click on it to access it with the DCV browser if it does not start automatically	
Open a console Menu Applications -> System Tools -> Terminal	
Enjoy	
Once you're finished with your session, Go back to the web interface and delete your session "X" button on the session representation's top-right corner	

Example using Paraview



Example using Paraview

The screenshot shows the NICE EnginFrame Views web application running in Mozilla Firefox. The browser's address bar displays the URL: <https://ruche.mesocentre.universite-paris-saclay.fr:8443/enginframe/vdi/vdi.xml?uri=com.enginframe.inte>. The application interface includes a top navigation bar with links like 'Mésocentre Moulon', 'User documentation', 'Accueil | MYCS', 'start [Wiki de fusion]', 'Annuaire', 'Projects - Dashboard...', 'Home - Wiki - mésoc...', and 'NICE EnginFrame Vi...'. A user login 'Welcome, josling' is visible in the top right. The main content area is titled 'Sessions' and features a 'Refresh' button, a search bar 'Search by session name', and a 'Sort by: Started on' dropdown. A yellow box in the center of the sessions list states 'No records to view'. On the left, a 'Menu' sidebar lists 'Monitor', 'Sessions' (highlighted), 'Hosts', and 'Services' (with sub-items for Linux). On the right, a 'FILTERS' sidebar shows 'All', 'Active' (highlighted), 'Starred', and 'Created Today'. The footer contains the NICE logo, the website www.nice-software.com, and a disclaimer: 'All trademarks and logos on this page are owned by NICE s.r.l. or by their respective owners. [Legal Notices](#)'.

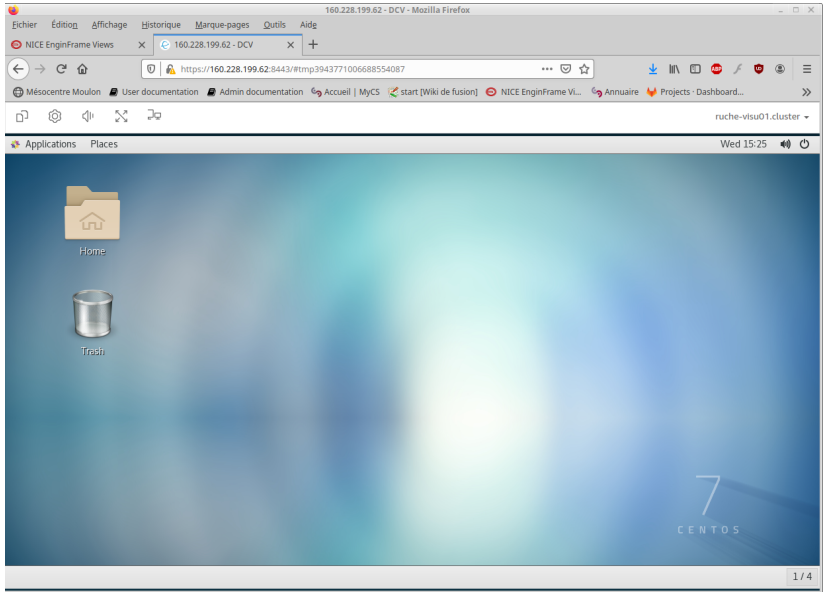
Example using Paraview

The screenshot shows a web browser window titled "NICE EnginFrame Views - Mozilla Firefox". The address bar displays the URL: https://ruche.mesocentre.universite-paris-saclay.fr:8443/enginframe/vdi/vdi.xml?_service=interactive_a8c71f6. The browser's menu bar includes "Fichier", "Édition", "Affichage", "Historique", "Marque-pages", "Outils", and "Aide". The page header features the "ruche" logo, a navigation menu with links like "Mésocentre Moulon", "User documentation", "Accueil", "MYCS", "start [Wiki de fusion]", "Annuaire", "Projects - Dashboard...", "Home - Wiki - mésoc...", and "NICE EnginFrame Vi...", and a user status bar showing "Welcome, josling" with links for "Settings" and "Logout".

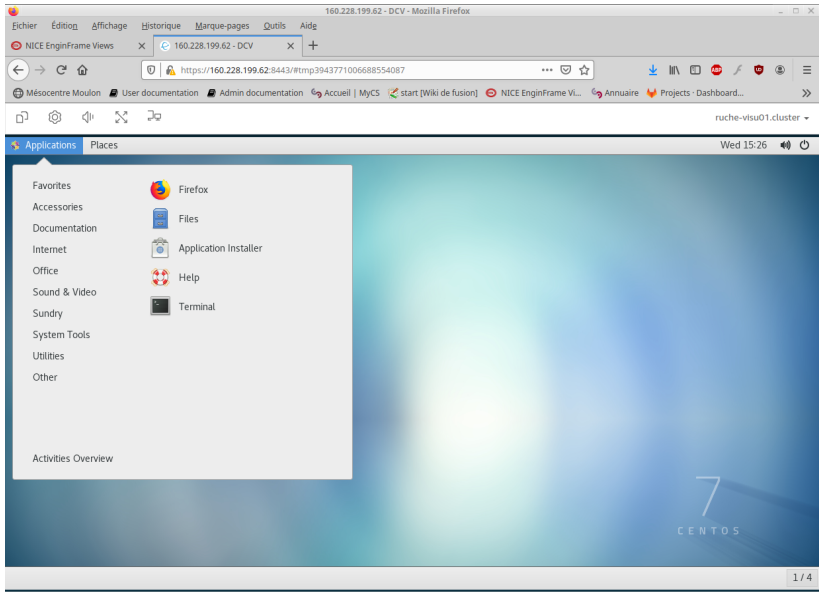
The main content area is titled "Linux (select resources)" and contains a sidebar menu on the left with the following items: "Monitor", "Sessions", "Hosts", "Services", "Linux (2h, 1cpu, ...)", and "Linux (select res...)". The "Linux (select res...)" item is highlighted. The main panel displays configuration options for a session: "Walltime : 1h", "Ncpus : 1", and "Mem : 4 gb", each with a dropdown arrow. A "Launch Session" button is positioned below these settings.

The footer of the page includes the NICE logo, the website www.nice-software.com, and a disclaimer: "All trademarks and logos on this page are owned by NICE s.r.l. or by their respective owners." A link to "Legal Notices" is also provided.

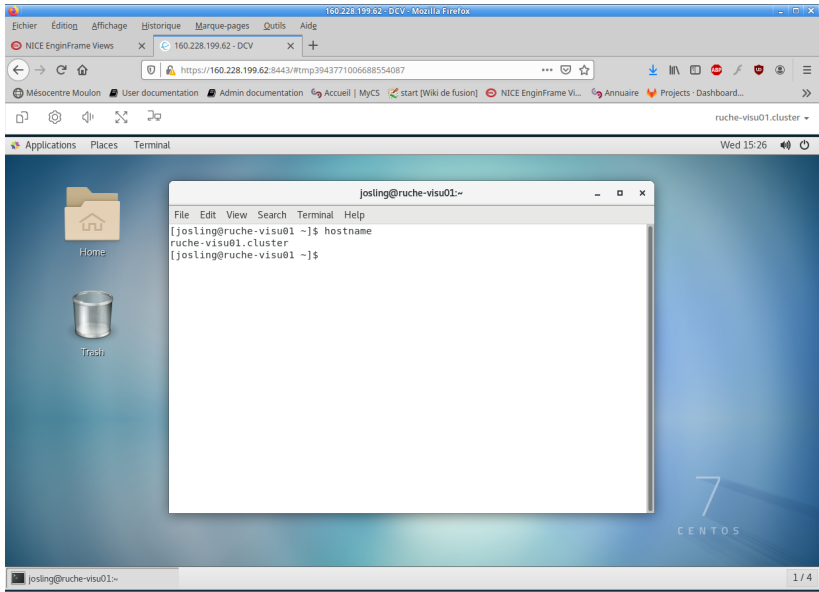
Example using Paraview



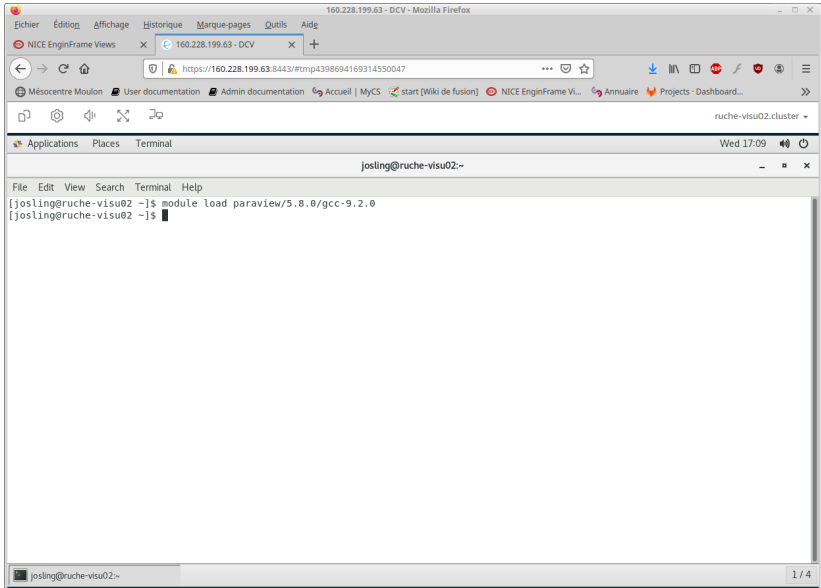
Example using Paraview



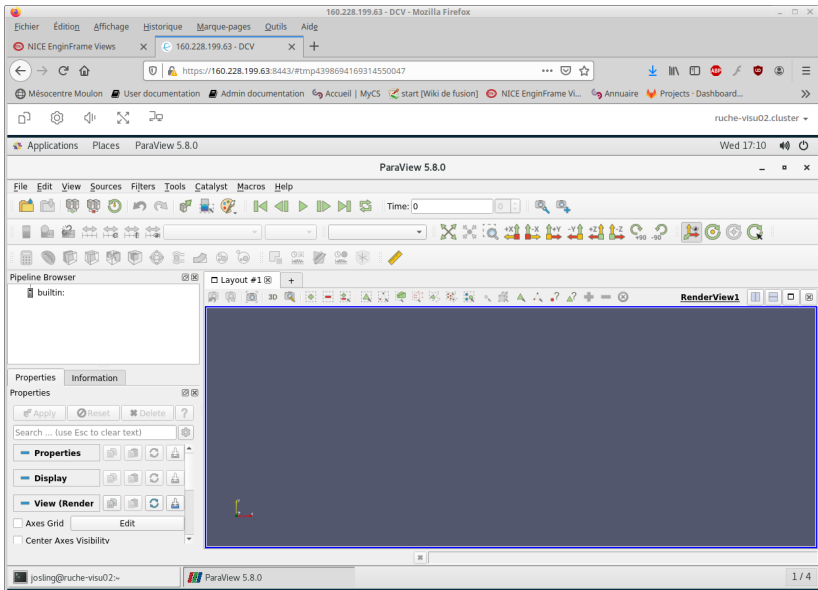
Example using Paraview



Example using Paraview



Example using Paraview



Example using Paraview

